

Neuroimaging Data Collection with fNIRS and EEG

Yiu Hong Ko (30-Jul-2026)

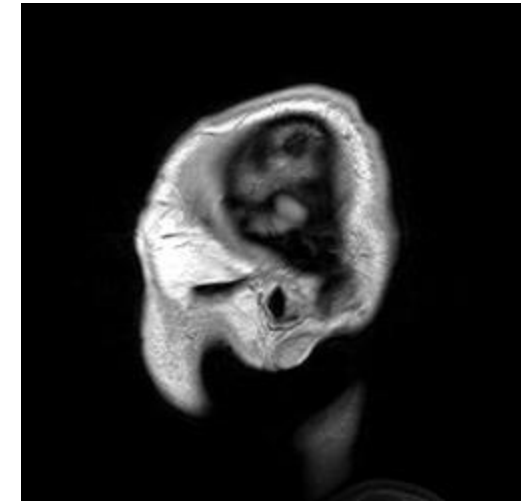
Outline

- What is neuroimaging?
- Functional near infrared spectroscopy (fNIRS)
 - brief history
 - mechanism
 - data collection demonstration
- Electroencephalography (EEG)
 - brief history
 - mechanism
 - data collection demonstration

What is neuroimaging?

Neuroimaging

- "Neuroimaging is the discipline that deals with the in vivo depiction of anatomy and function of the central nervous system (CNS) in health and disease." (Fulham, 2004)
- Used in disciplines such as:
 - Radiology
 - Neurology
 - Psychology
 - Education
 - Social science
 - Linguistics
 - Economy
 - Law



<https://en.wikipedia.org/wiki/Neuroimaging>

Neuroimaging

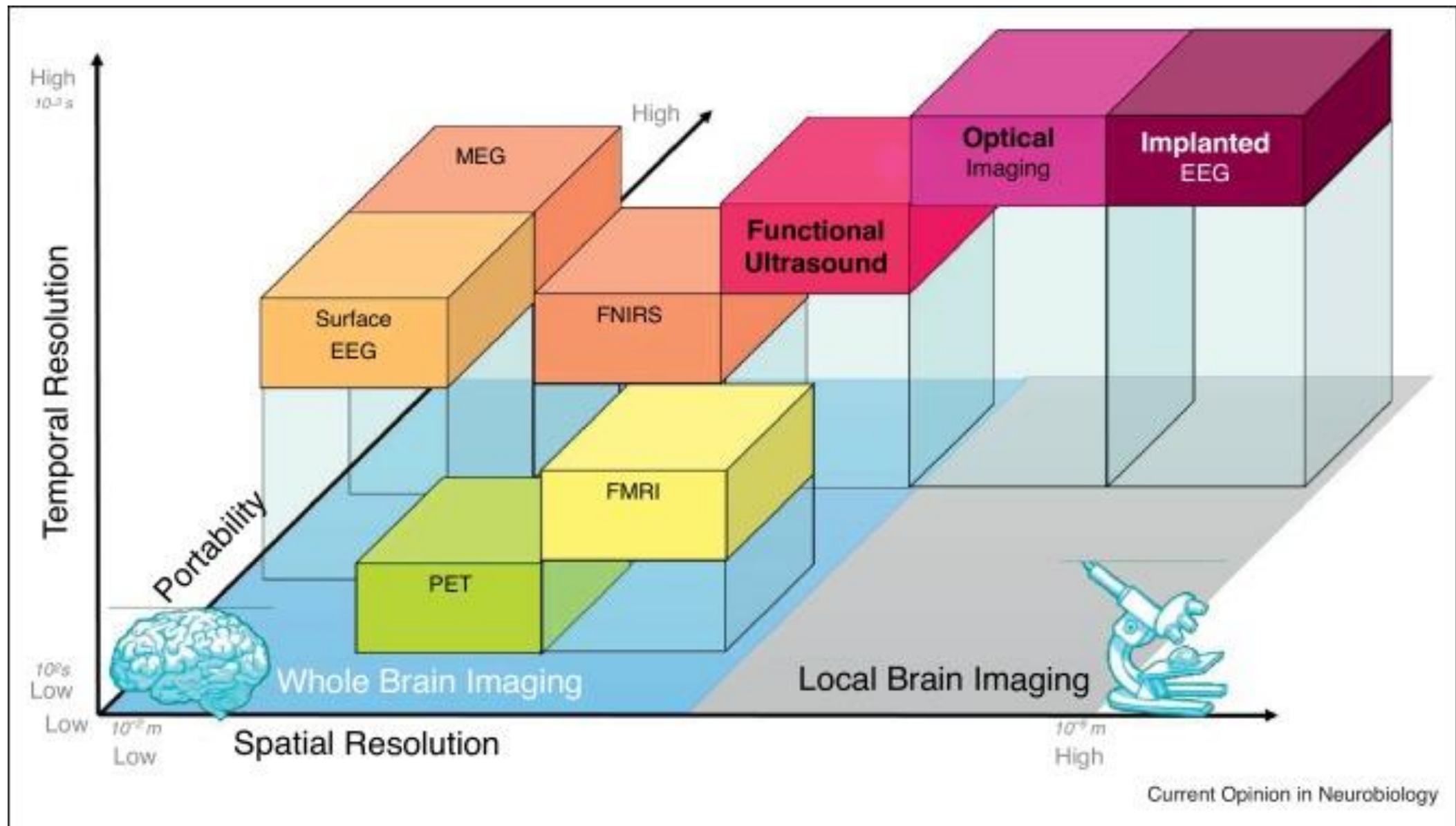
- Structural neuroimaging (anatomy of the brain)
 - Magnetic resonance imaging (MRI)
- Hemodynamic neuroimaging
 - Functional magnetic resonance imaging (fMRI)
 - Functional near-infrared spectroscopy (fNIRS)
- Electrophysiological neuroimaging
 - Electroencephalography (EEG)
 - Magnetoencephalography (MEG)



https://en.wikipedia.org/wiki/Magnetic_resonance_imaging



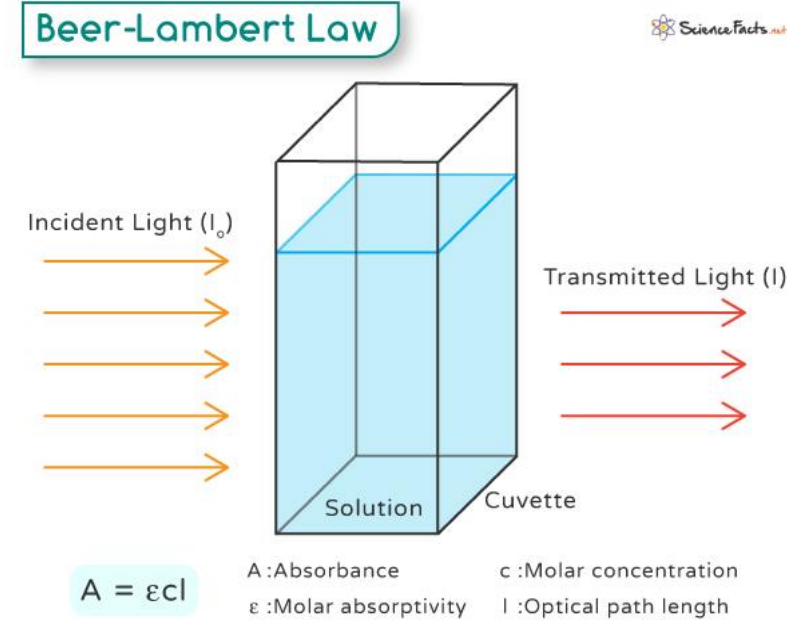
<https://en.wikipedia.org/wiki/Magnetoencephalography>



fNIRS

Brief history of fNIRS

- Bouguer (1729)
 - intensity of light decreases as it passes through a medium
- Lambert (1760)
 - light absorption is directly proportional to the thickness of the sample material
- Beer (1852)
 - light absorption is also directly proportional to the concentration of a solution



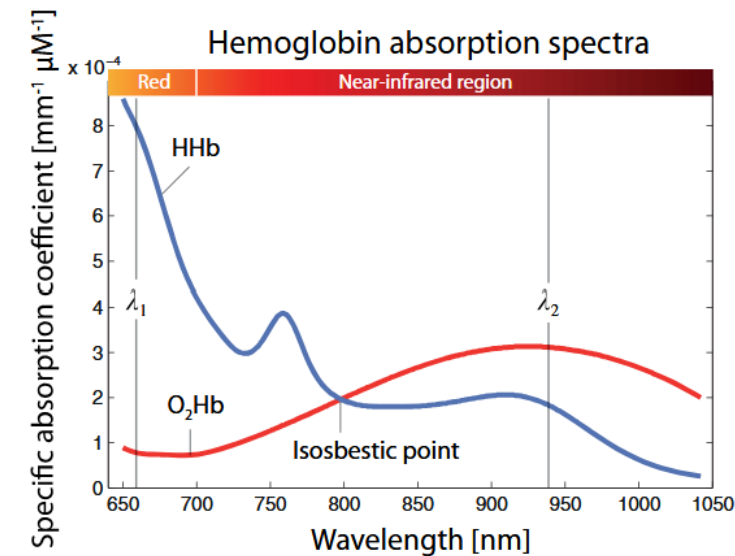
<https://www.sciencefacts.net/beer-lambert-law.html>

Brief history of fNIRS

- Hoppe-Seyler (1862)
 - "blood pigment" absorb light
- Stokes (1864)
 - haemoglobin (Hb) exists in two states (oxygenated vs. deoxygenated), which absorb light differently
- von Vierordt (1876)
 - in vivo demonstration of spectral differences
- Millikan (late 1933)
 - developed a colorimeter to measure oxygenation of Hb
- Matthes and Gross (1938)
 - first oximeter (similar to a pulse oximeter)

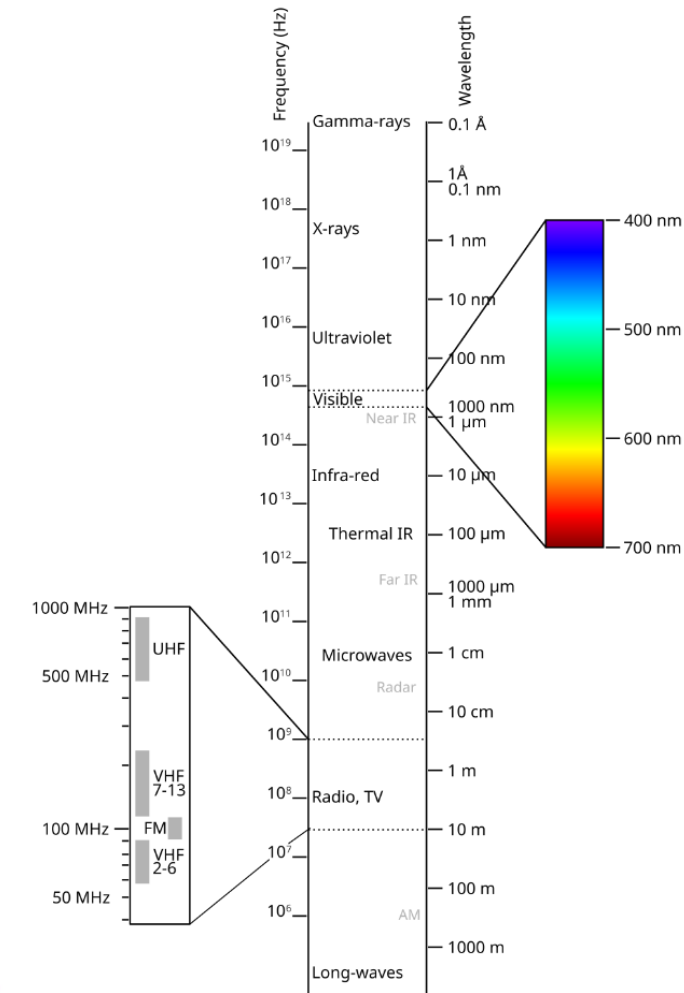
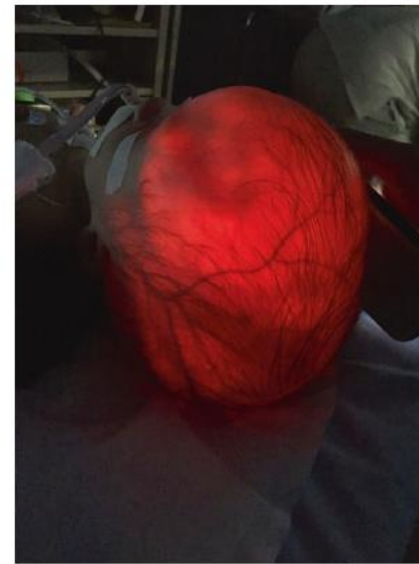


https://en.wikipedia.org/wiki/Functional_near-infrared_spectroscopy

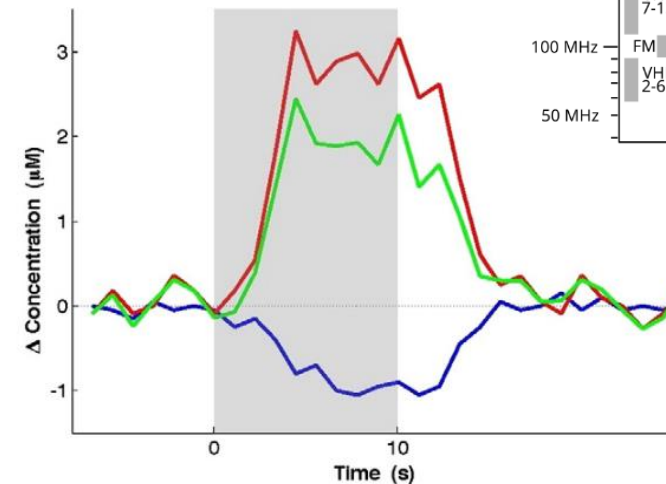


Brief history of fNIRS

- Bright (1831)
 - Transillumination of the neonatal skull
- Jöbsis (1977)
 - detect changes of adult cortical oxygenation during hyperventilation by near-infrared spectroscopy
- Hoshi and Tamura (1992)
 - first fNIRS human study: prefrontal cortex oxygenation changed during a mental task



Oxygenated
Deoxygenated
Total



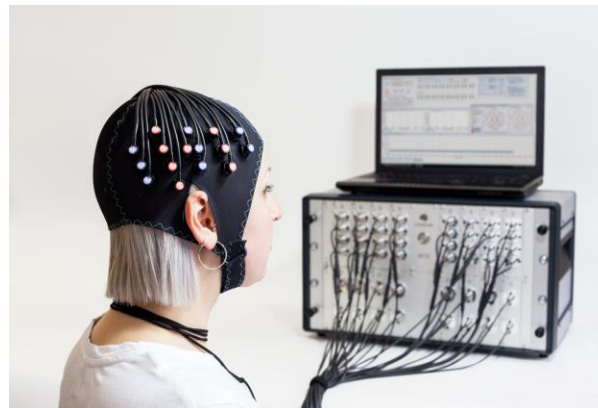
Ferrari, M., & Quaresima, V. (2012). A brief review on the history of human functional near-infrared spectroscopy (fNIRS) development and fields of application. *NeuroImage*, 63(2), 921–930.

<https://doi.org/10.1016/j.neuroimage.2012.03.049>

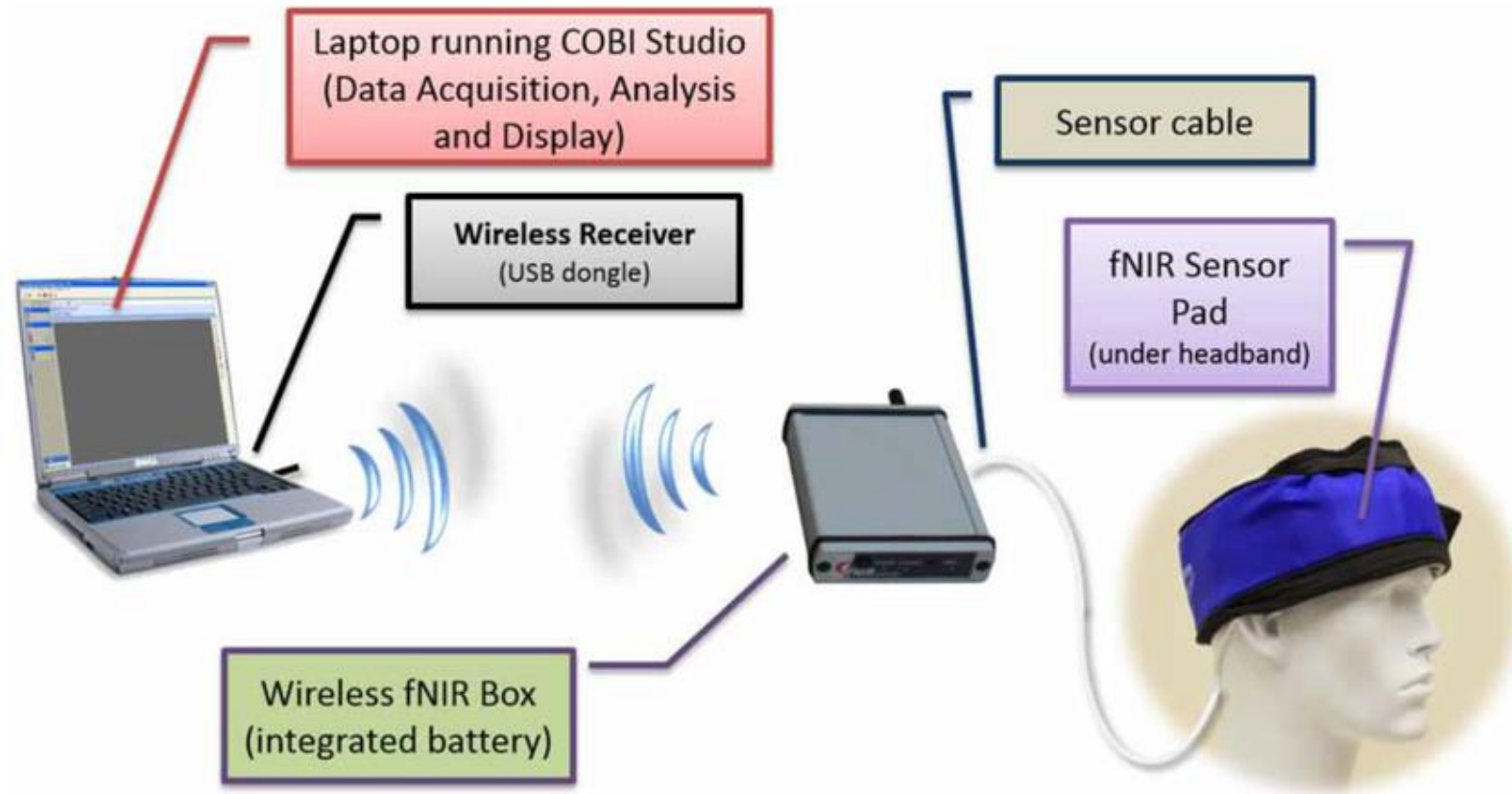
Joaquim, A. F., Ghizoni, E., Tedeschi, H., & Ferreira, M. A. T. (Eds.). (2019). *Fundamentals of Neurosurgery: A Guide for Clinicians and Medical Students*. Springer International Publishing.

<https://doi.org/10.1007/978-3-030-17649-5>

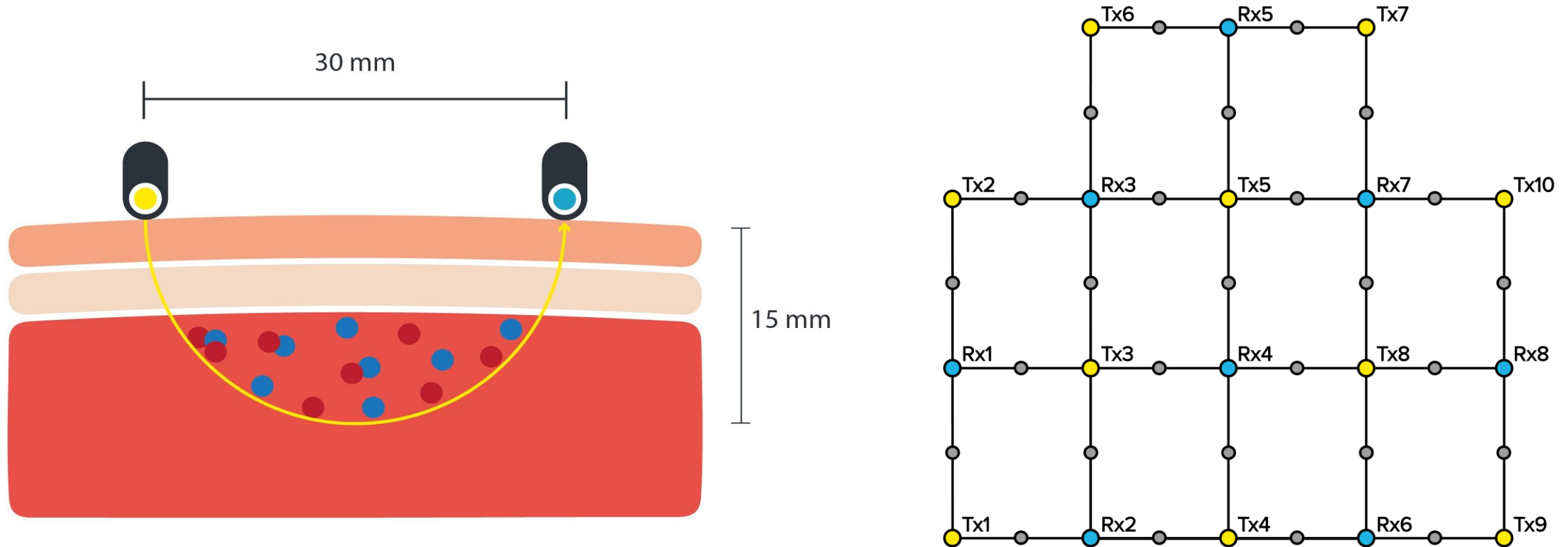
fNIRS systems



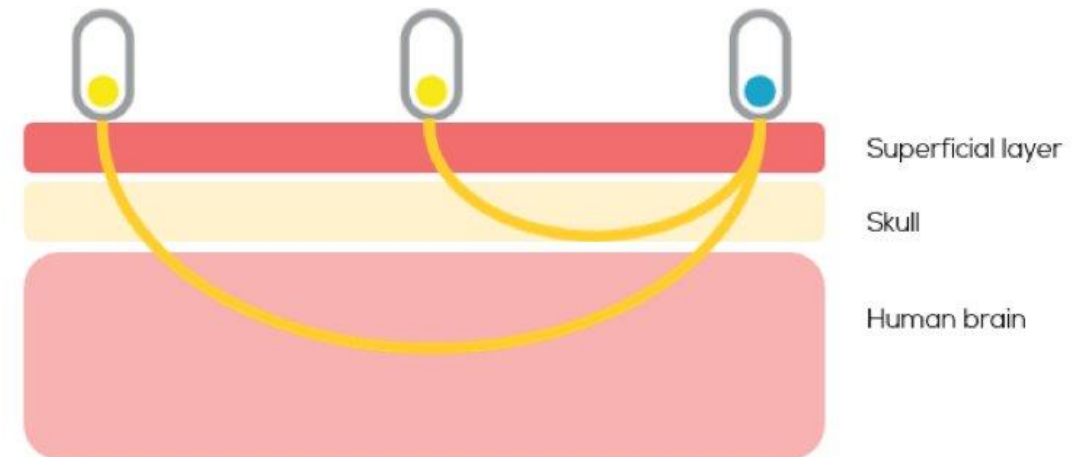
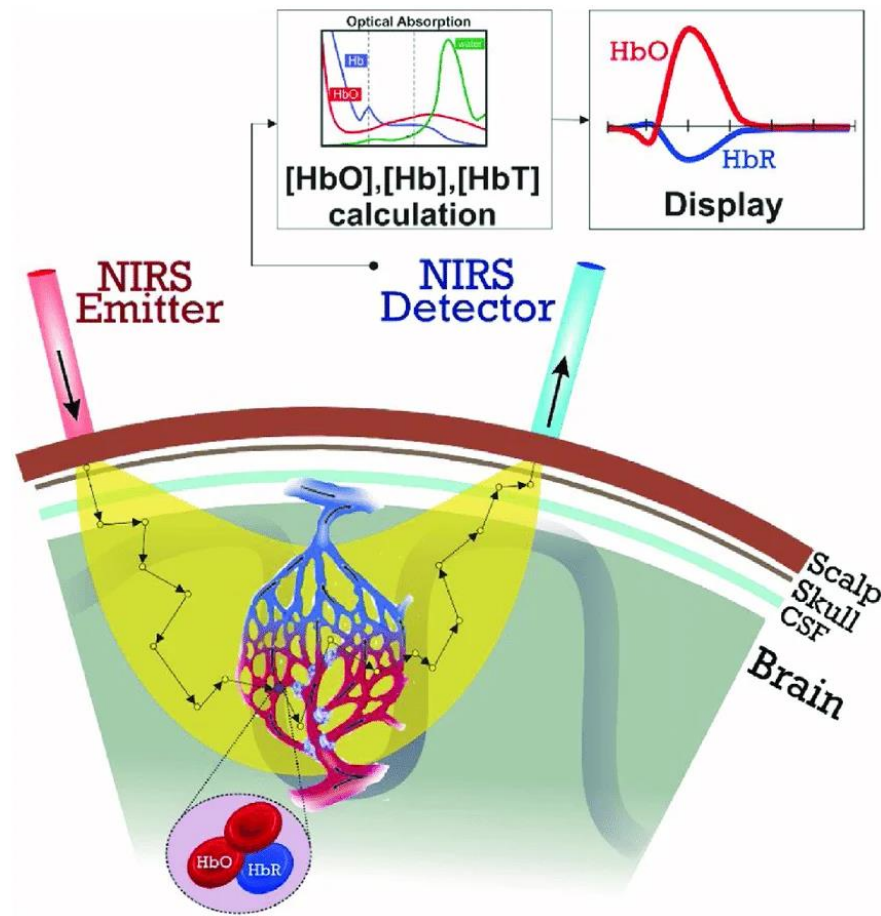
Components of an fNIRS system



Components of an fNIRS system



Mechanism of fNIRS



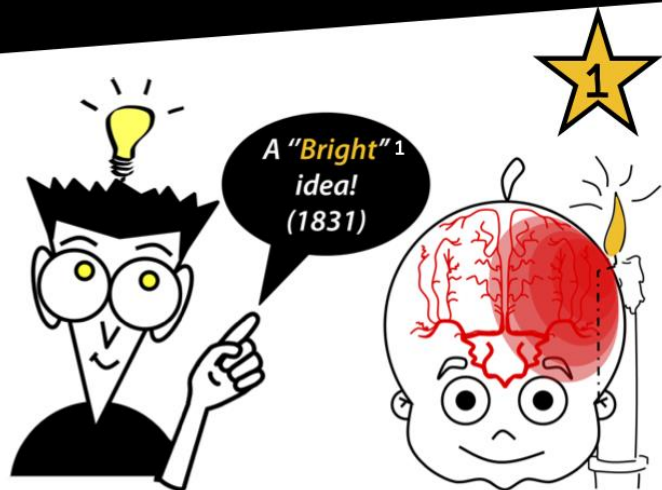
<https://artinis.com/blogpost-all/2019/2/6/short-separation-channels-the-new-trend-in-fnirs>

Mohammadi-Nejad, A.-R., Mahmoudzadeh, M., Hassanpour, M. S., Wallois, F., Muzik, O., Papadelis, C., Hansen, A., Soltanian-Zadeh, H., Gelovani, J., & Nasiriavanaki, M. (2018). Neonatal brain resting-state functional connectivity imaging modalities. *Photoacoustics*, 10, 1–19. <https://doi.org/10.1016/j.pacs.2018.01.003>

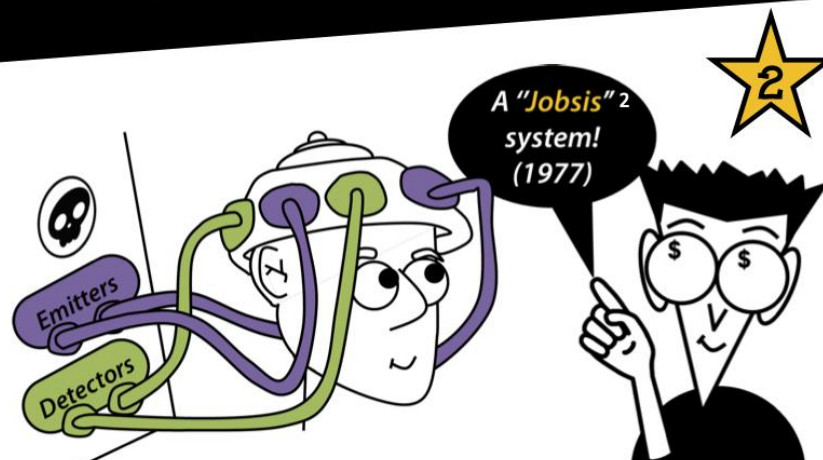
A fun and easy way
to see the brain at work

Functional Near InfraRed Spectroscopy (fNIRS) NeuroImaging Method to Monitor Cortical Activity in situ FOR DUMMIES

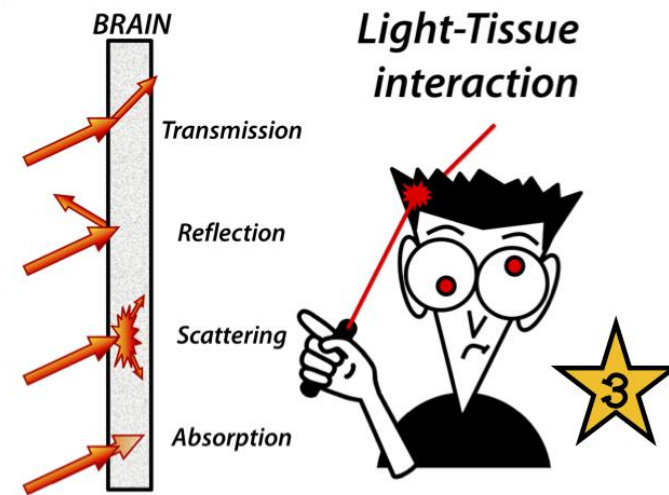
Follow the ★



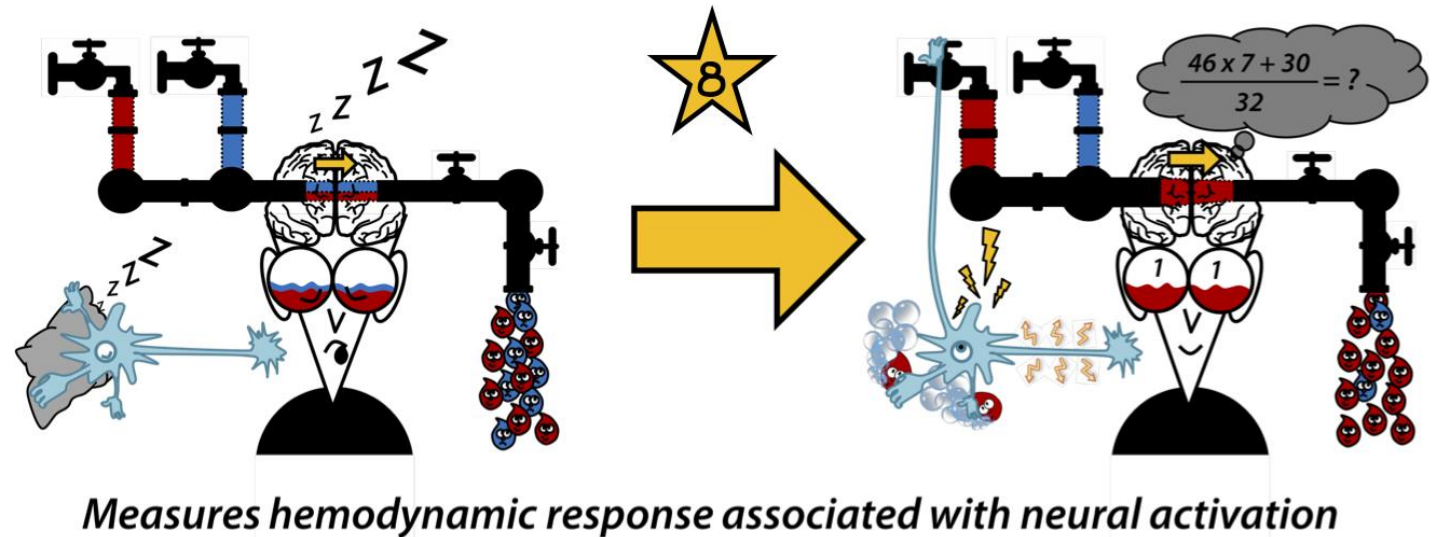
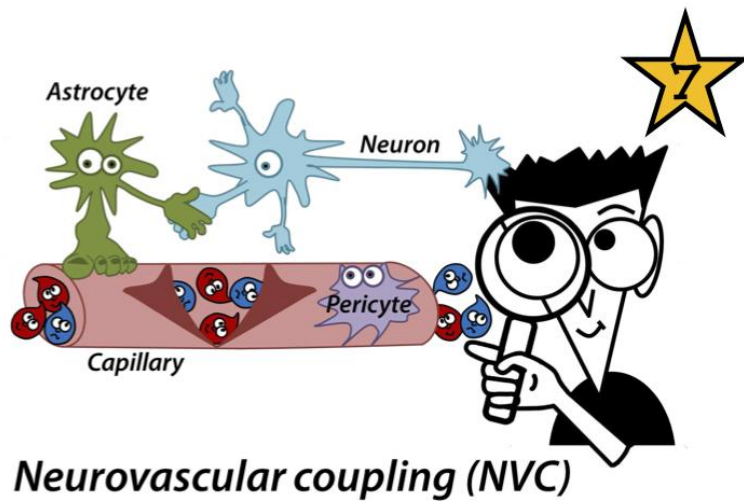
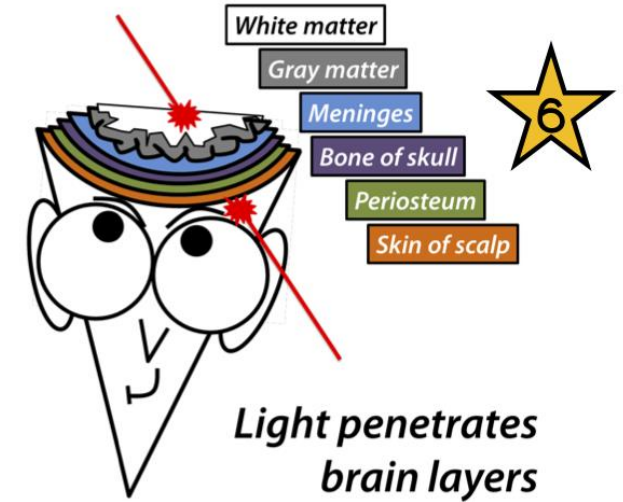
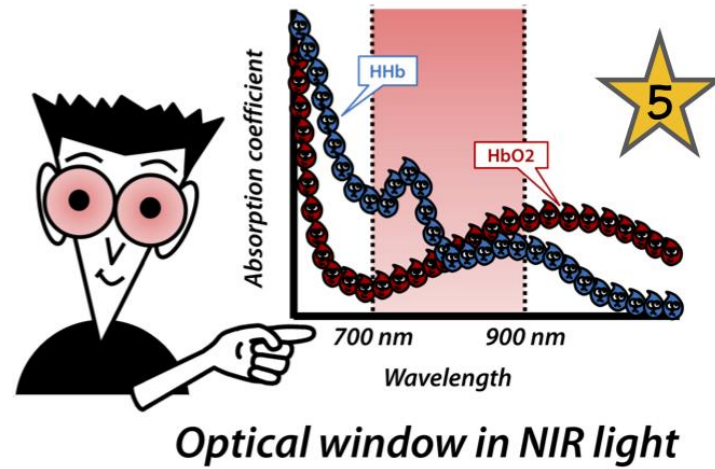
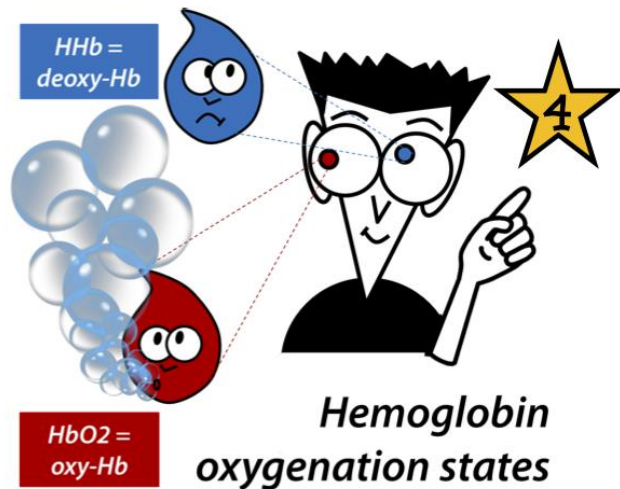
Transillumination of the neonatal skull: seeing the light

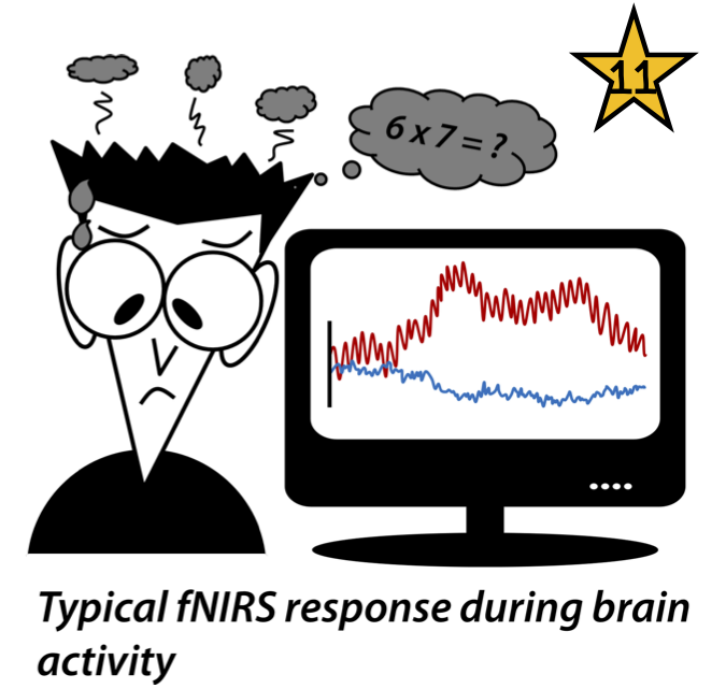
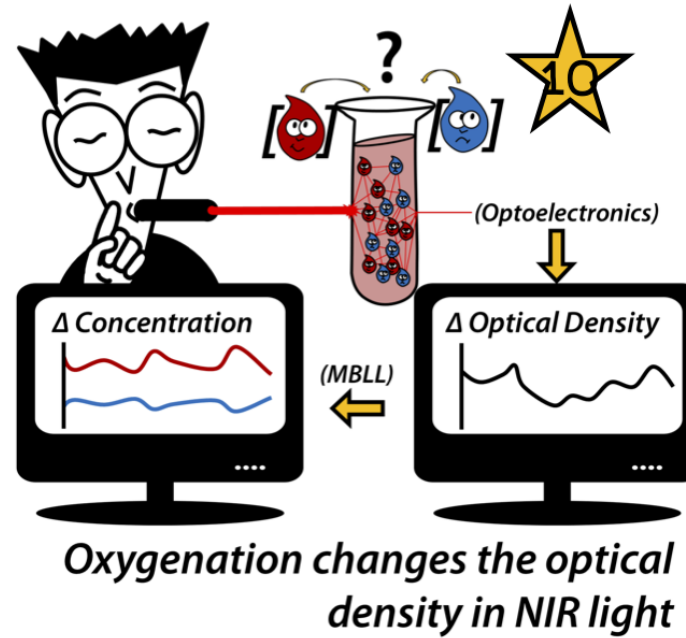
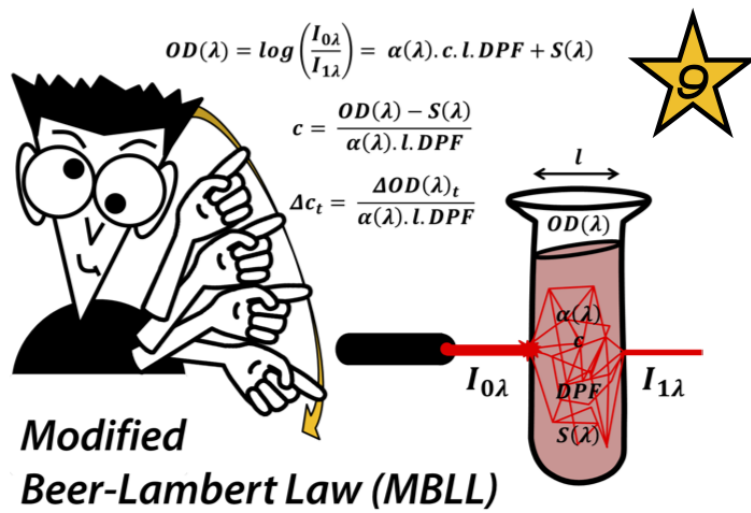


Noninvasive monitoring of cerebral oxygenation

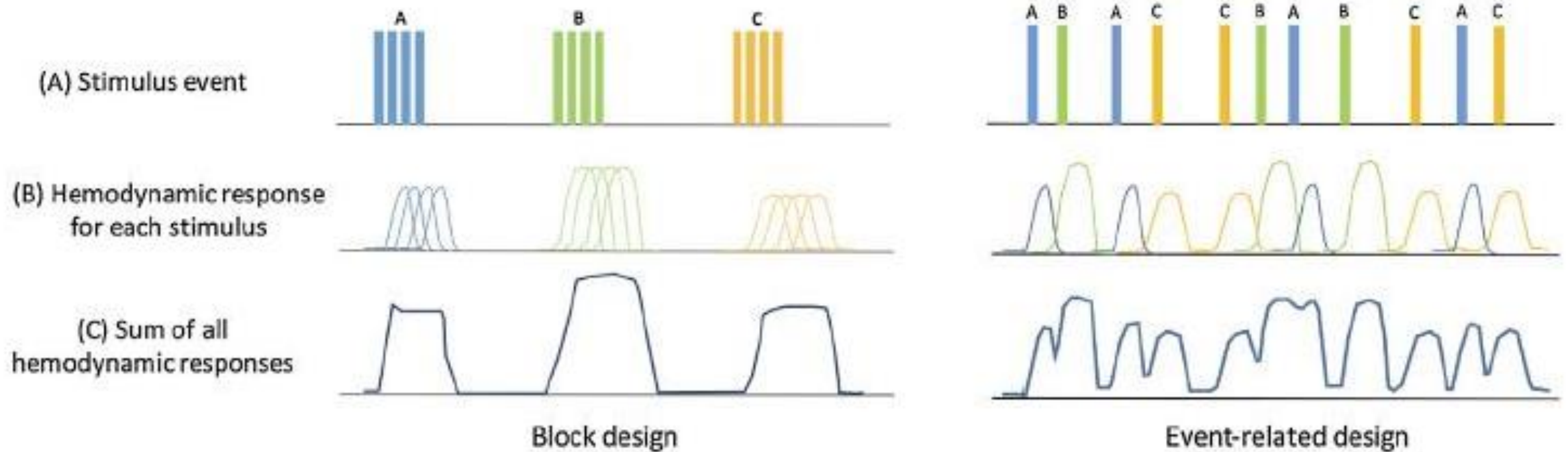


Light-Tissue interaction

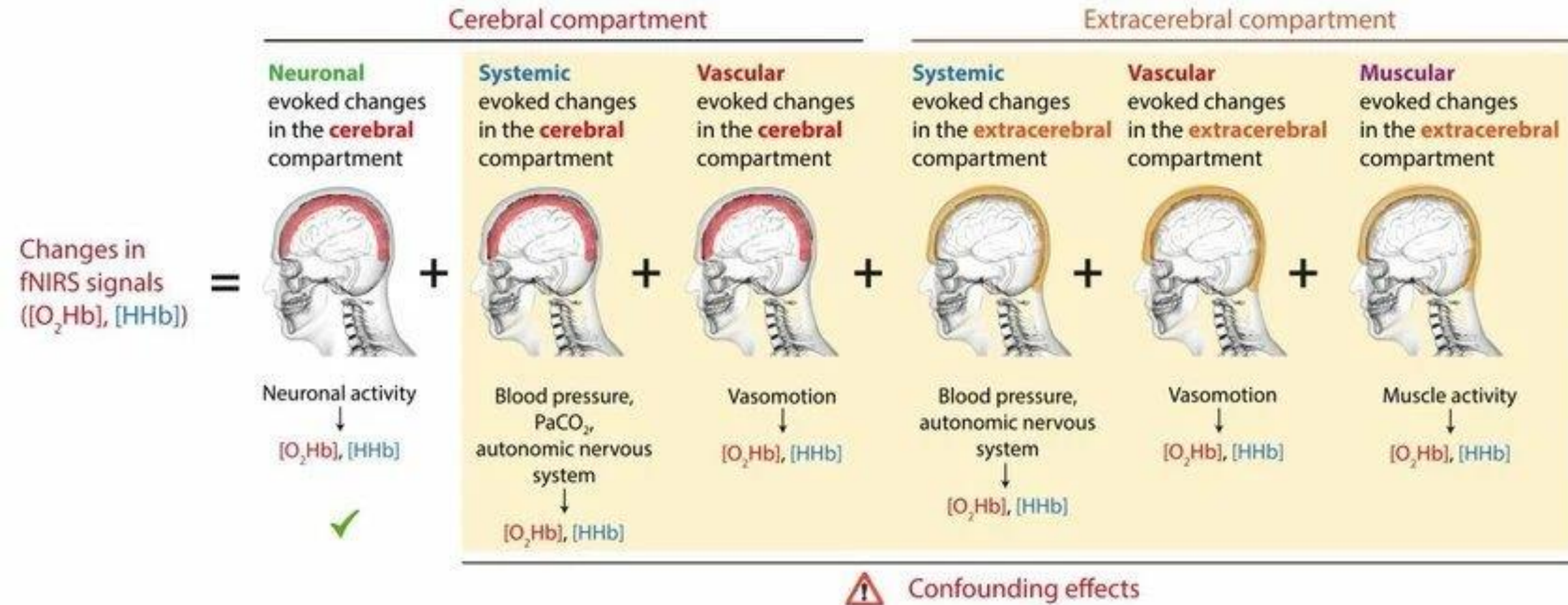




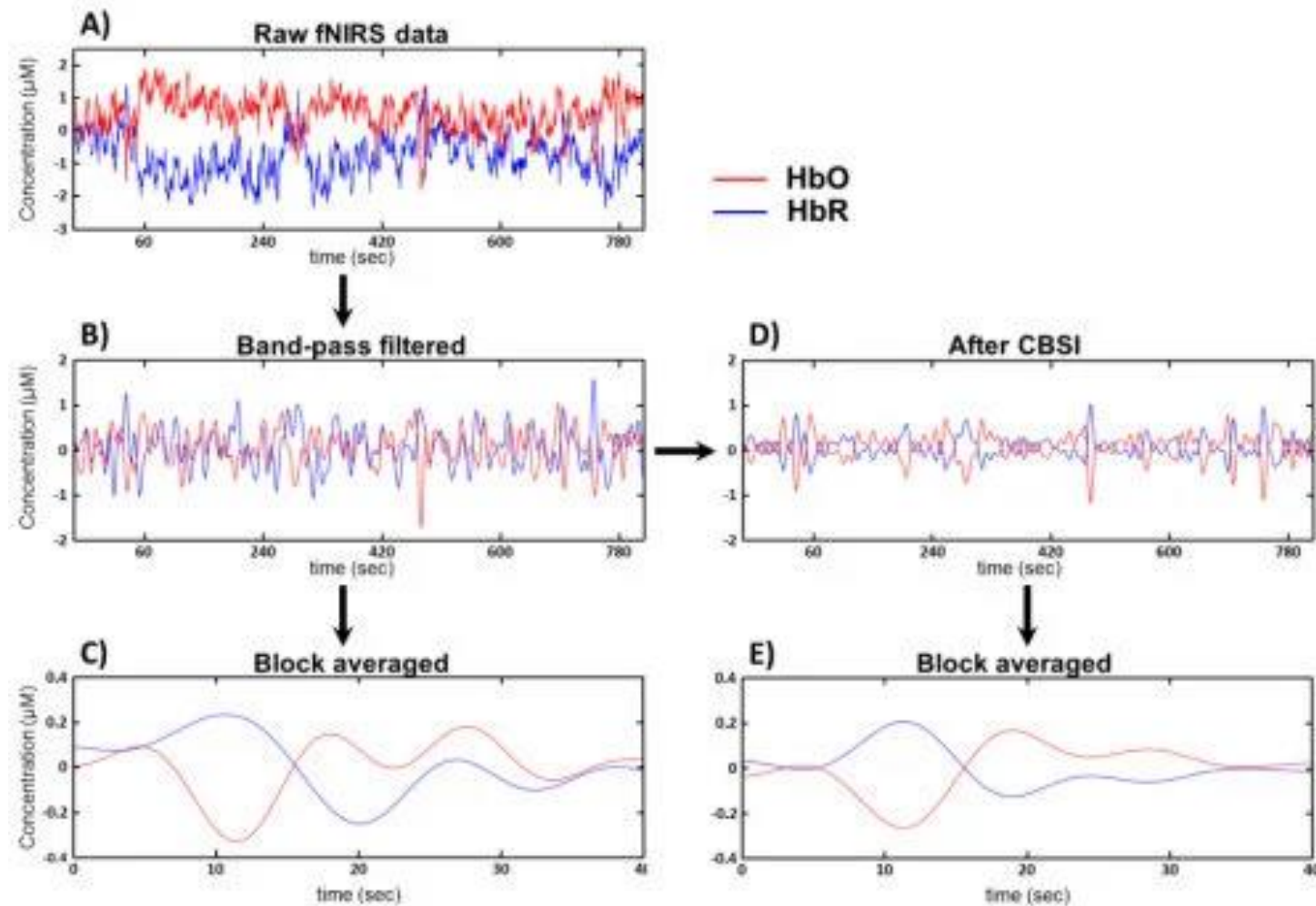
fNIRS data – experimental design



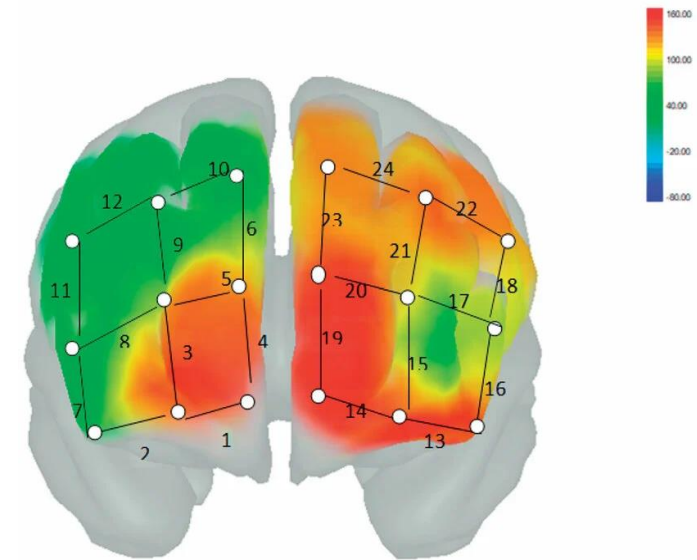
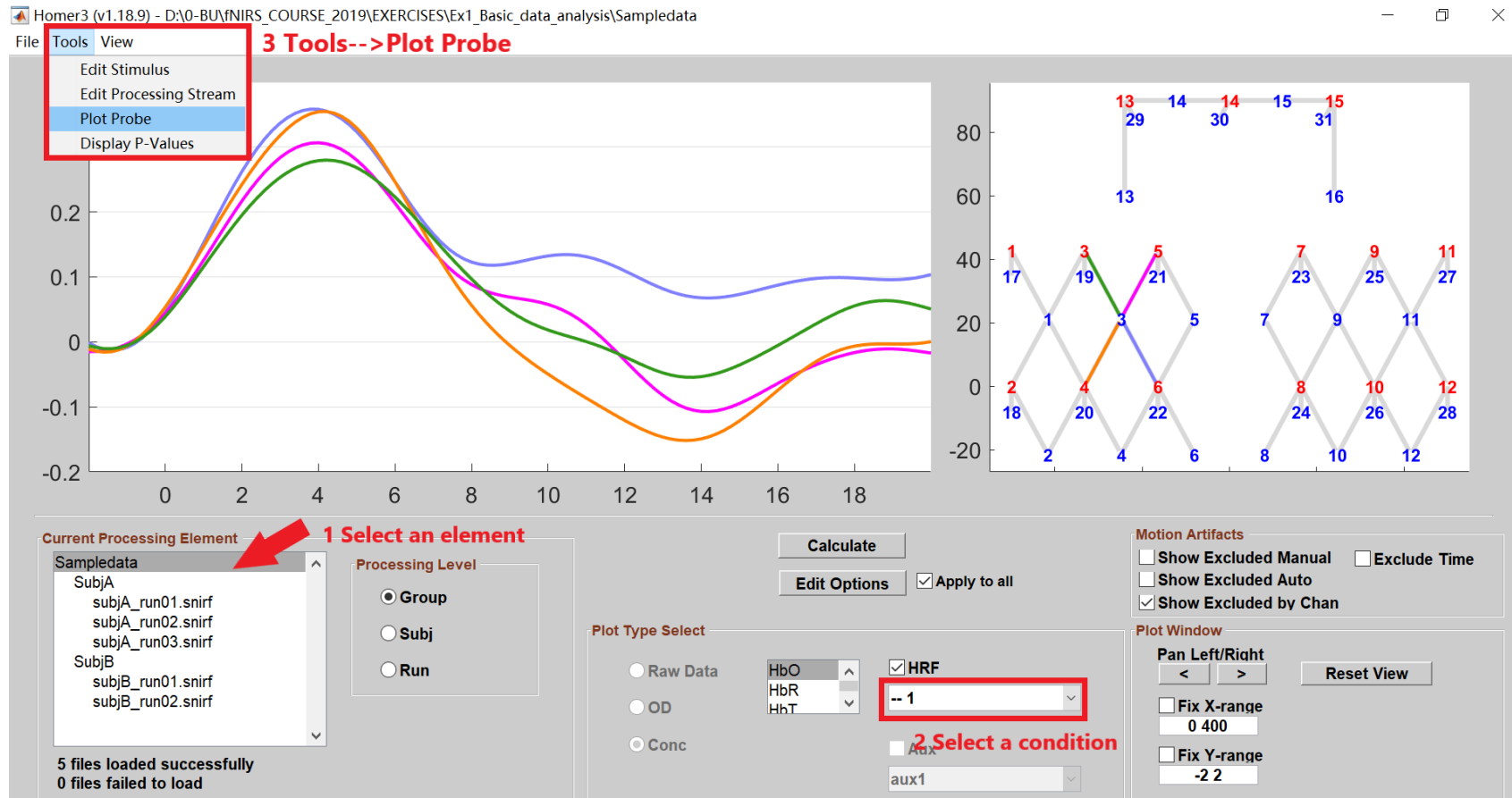
fNIRS data - artefacts



fNIRS data – processing



fNIRS data - data processing



<https://github.com/BUNPC/Homer3/wiki/Viewing-HRF-in-Probe-Space>

Khandekar, P., Shenoy, S., & Sathe, A. (2023). Prefrontal cortex hemodynamic response to acute high intensity intermittent exercise during executive function processing. *The Journal of General Psychology*, 150(3), 295–322. <https://doi.org/10.1080/00221309.2022.2048785>

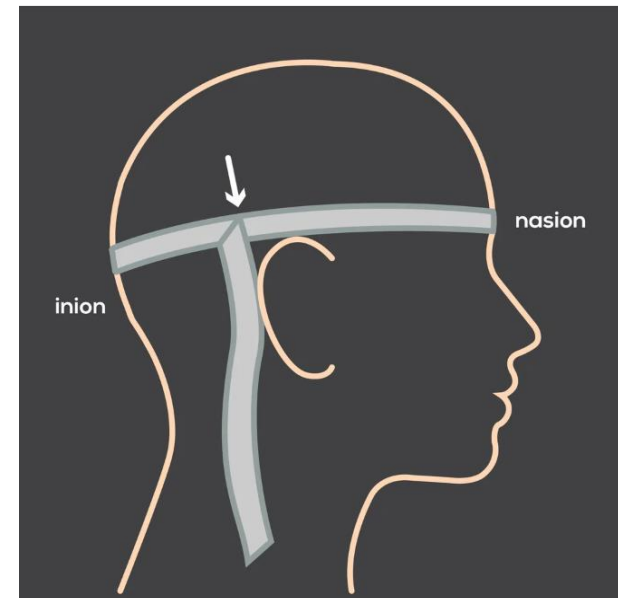
Applications of fNIRS

- Neurology
 - neurological disorders, e.g., Alzheimer's Disease
- Psychiatry
 - mental disorders, e.g., depression
- Psychology/education
 - e.g., attention

Guidelines on fNIRS data collection – task-based measure (Artinis Brite MKIII)

1. Measure head size
2. Setup optode configuration
3. Help the participant wear the cap
4. Make sure that the cap is centered, and the distance between the cap and eyebrows should be one tenth of the distance between the eyebrows to inion.

Size	Circumference
S	53-55cm
M	55-57cm
L	57-59cm
XL	59-61cm

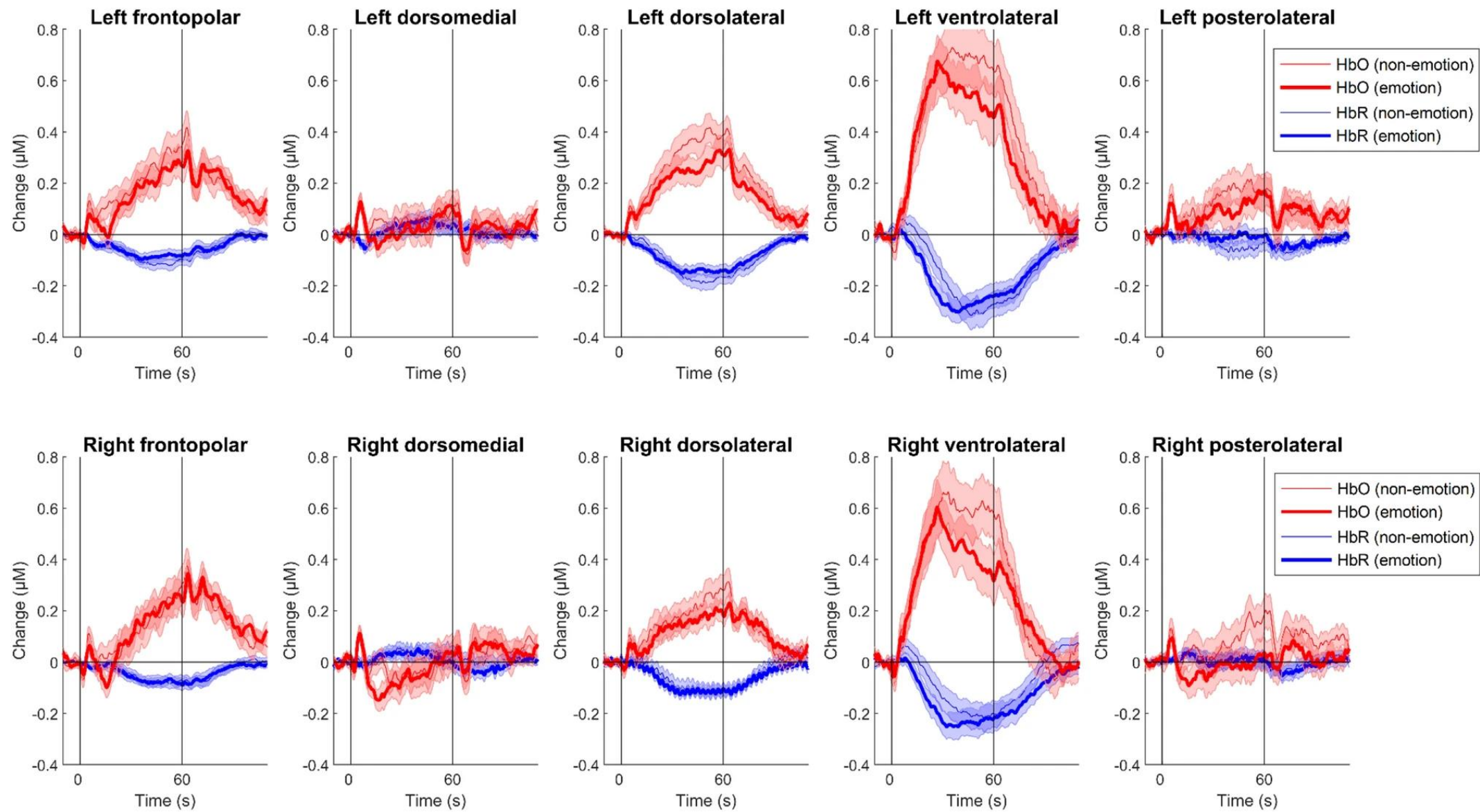


Guidelines on fNIRS data collection – task-based measure (Artinis Brite MKIII)

5. Open computer task program (e.g., PsychoPy)
6. Open the data collection software
7. Movement calibration
8. Check signal quality, adjust optodes
9. Start recording

Verbal fluency test (Semantic fluency test)

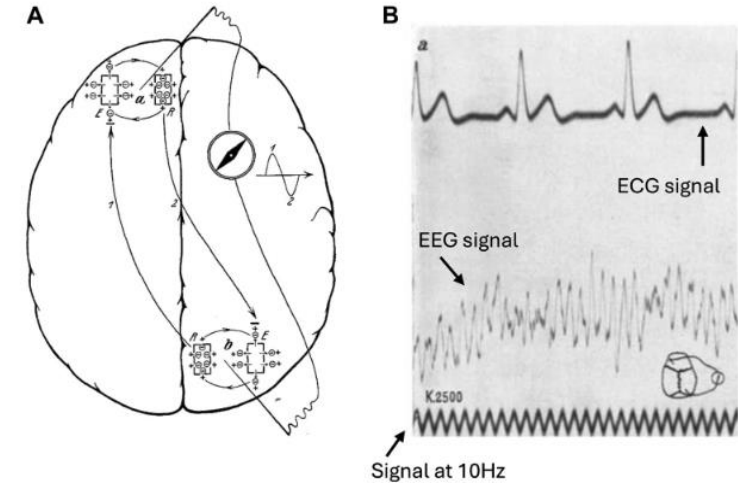
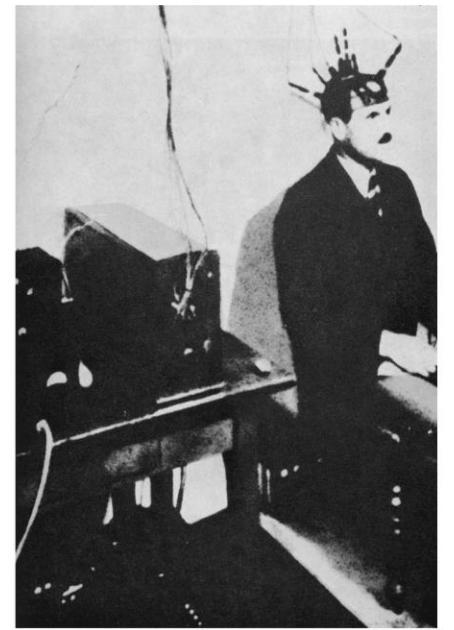
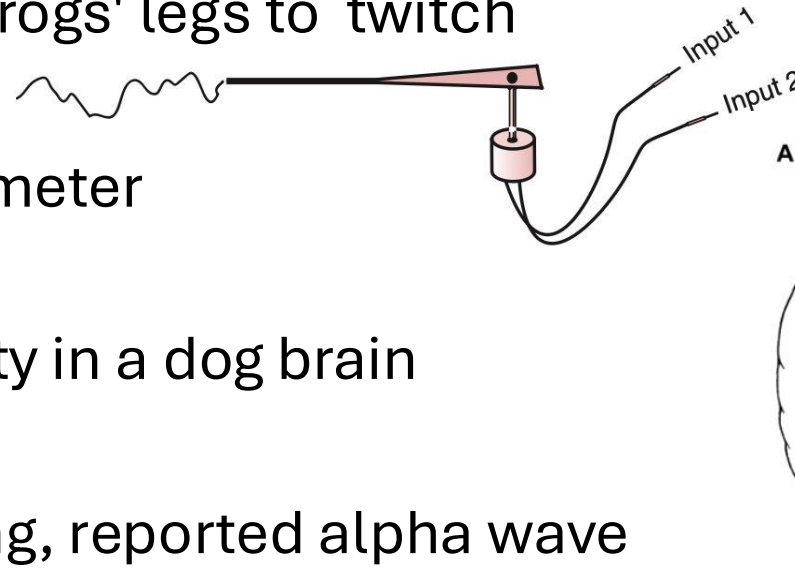
- Participants were asked to generate as many unique words as possible from a specific semantic category within 60 s for four abstract categories (country, occupation, positive emotion, negative emotion).



EEG

Brief history of EEG

- Newton (1717)
 - Proposed that the brain controls muscles by electricity
- Galvani (1786)
 - Electrical shock caused frogs' legs to twitch
- Thompson (1867)
 - Siphon recorder/galvanometer
- Caton (1875)
 - recorded electrical activity in a dog brain
- Berger (1929)
 - First human EEG recording, reported alpha wave



Libenson, M. H. (2024). *Practical approach to electroencephalography E-Book*. Elsevier Health Sciences.

Stark, L., & Campbell, N. D. (2018). The ineffable: A framework for the study of methods through the case of mid-century mind-brain sciences. *Social Studies of Science*, 48(6), 789–820. <https://doi.org/10.1177/0306312718816807>

Vergani, A. A. (2024). Hans Berger (1873–1941): The German psychiatrist who recorded the first electrical brain signal in humans 100 years ago. *Advances in Physiology Education*, 48(4), 878–881. <https://doi.org/10.1152/advan.00119.2024>

Brief history of EEG

- Davis and Forbes (1933)
 - Linked alpha wave with the relaxed state
- Gibbs and Lennox (1934)
 - Linked EEG signal with epilepsy (3/s spikes)



Absence seizure

The symptoms of absence seizures include:



Brief sudden staring spell or "blank stare."



Loss of awareness.



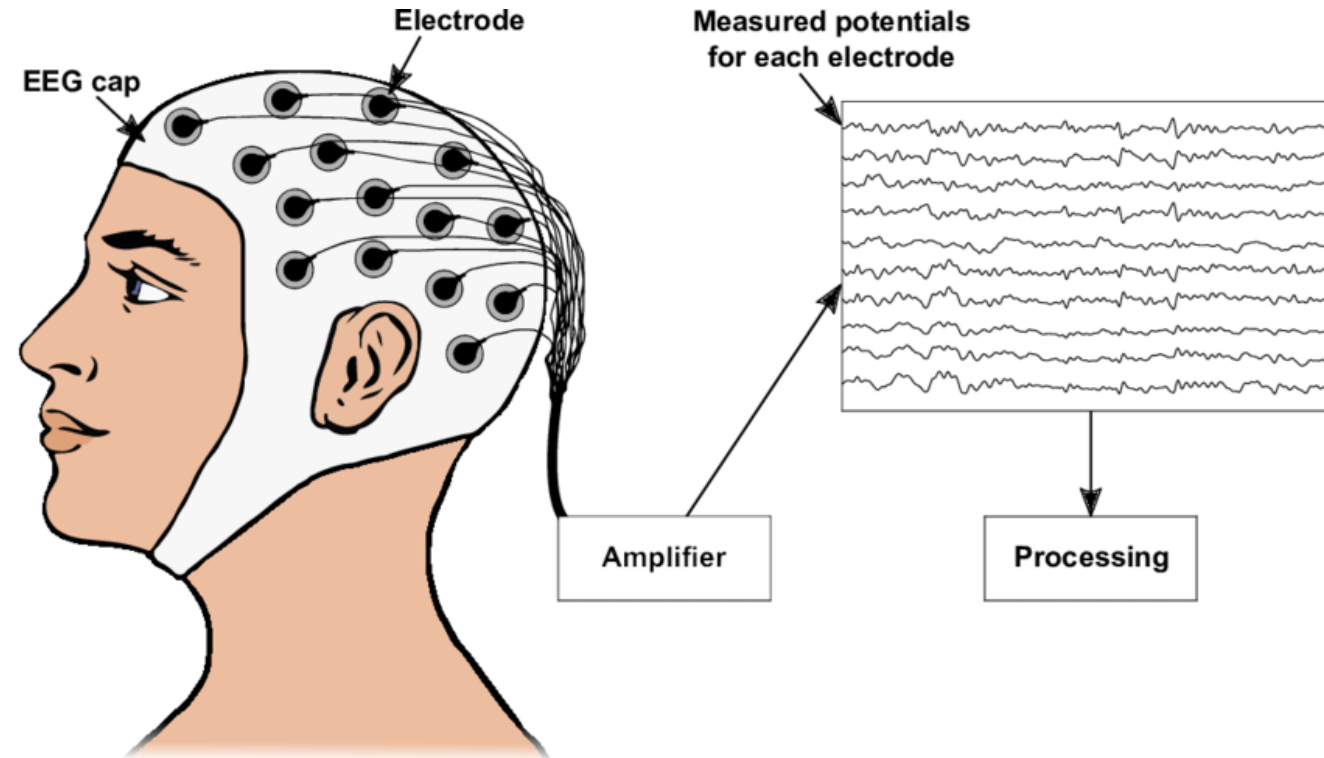
Complete stop in activity during the seizure.



Occasional nodding of the head.

Components of an EEG system

- Amplifier
- Computer(s)
- Electrodes
 - Scalp electrodes
 - Ground
(reduce external electrical noise)
 - External electrodes
 - Electrooculography (EOG)
 - Electrocardiography (ECG/EKG)
 - Reference
(reference for voltage at particulate electrode sites)



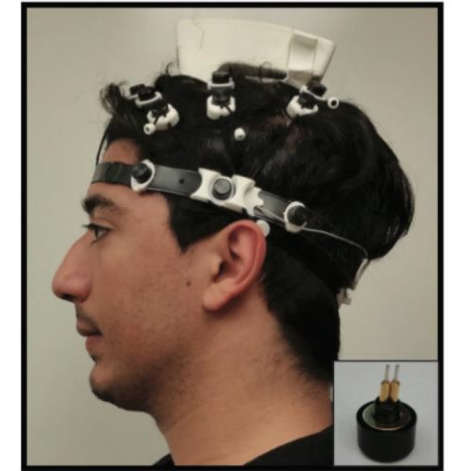
Scalp EEG systems

- Wet vs. dry
- Stationary vs mobile
- Passive vs. active electrodes

A. Biosemi Wet Electrodes

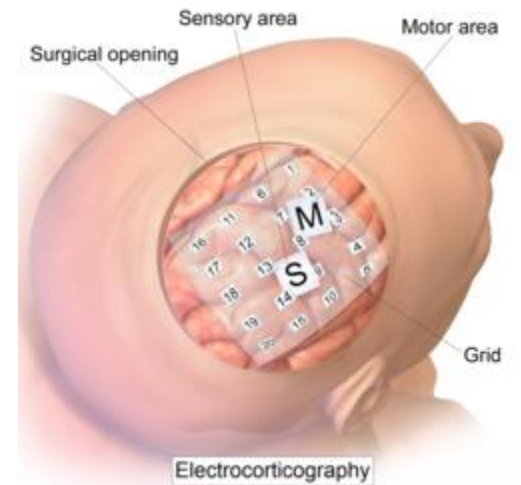
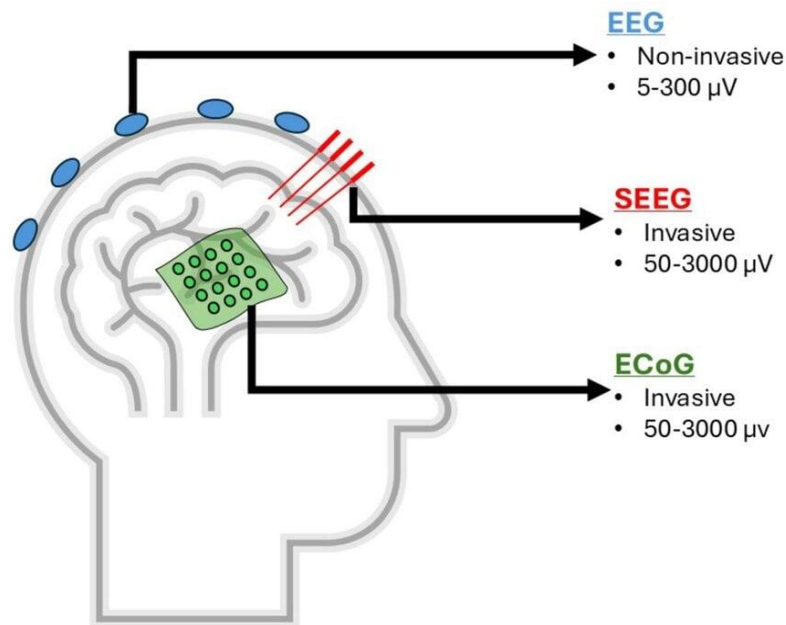


F1 Dry Electrodes



Other types of EEG

- Intracranial EEG (iEEG)
 - Stereotactic electroencephalograph (SEEG) with depth electrodes
 - Electrocorticogram (ECoG) with subdural electrode grids



<https://en.wikipedia.org/wiki/Electrocorticography>

International 10-20 electrode system

- Based on two fixed anatomical points, e.g., nasion and inion

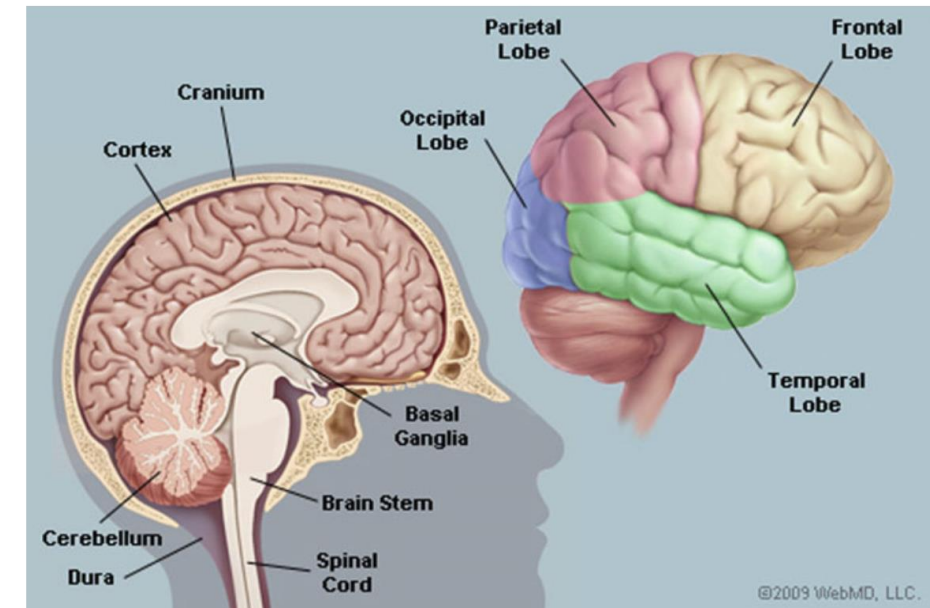
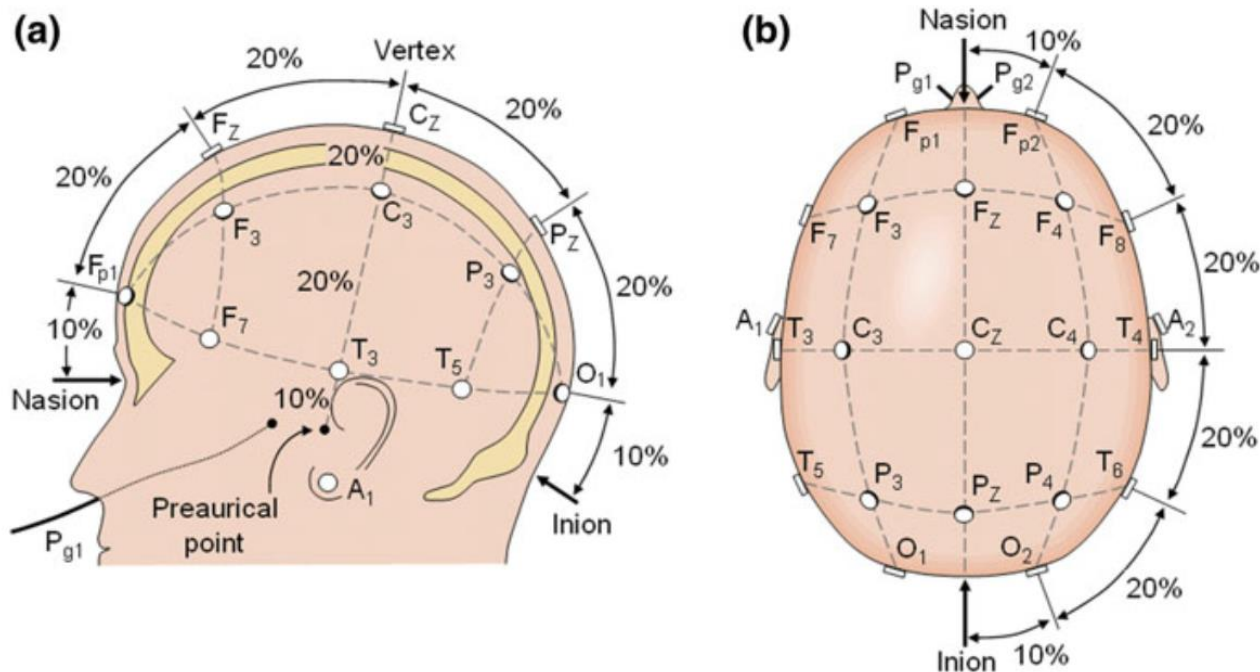
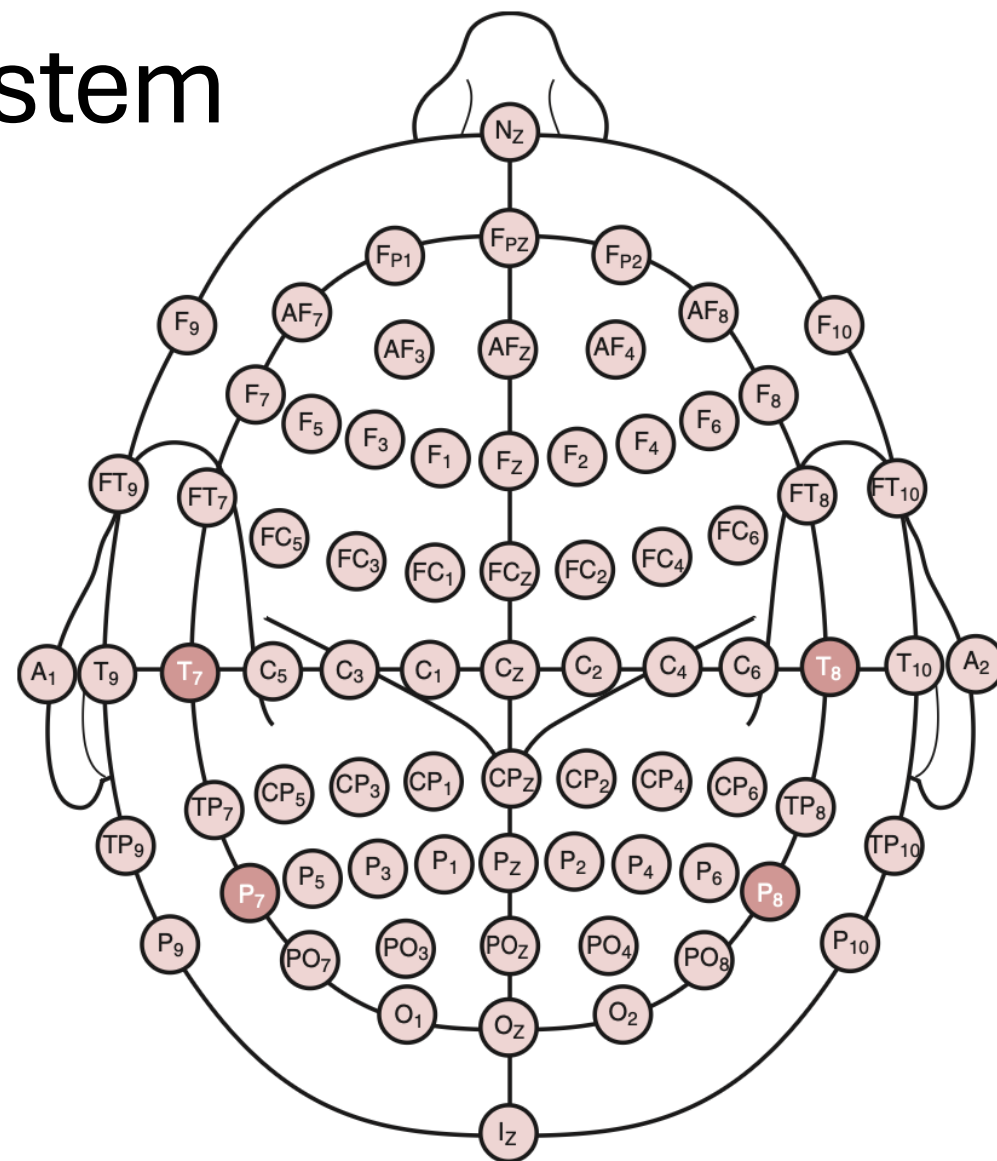


Fig. 1.10 Anatomical areas of the brain (Ref. Brain & Nervous System Health Center, WebMD)

Fig. 1.2 The international 10–20 electrode placement system (Campisi 2012; Jasper 1958; 10/20 positioning/DIY tDCS)

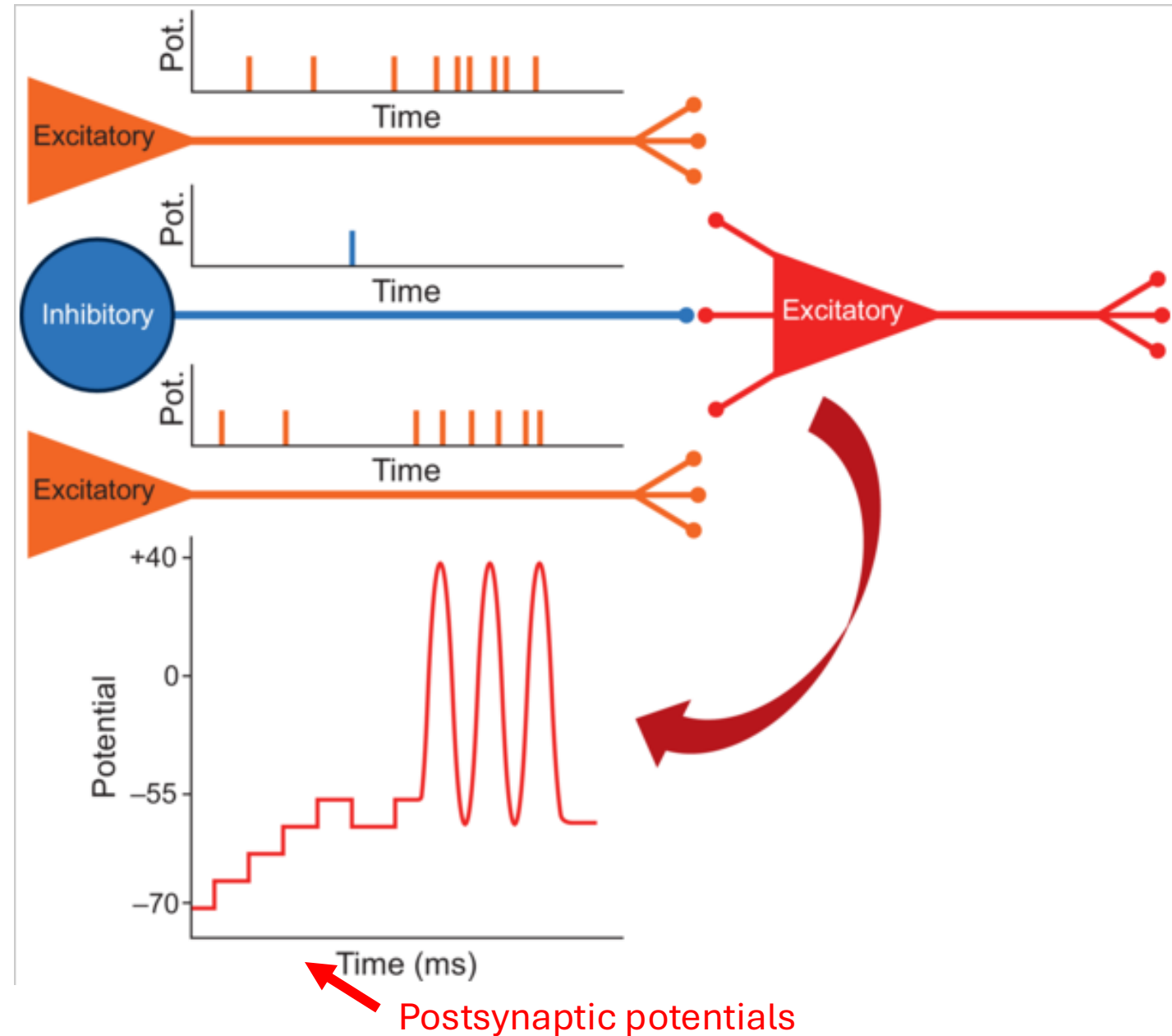
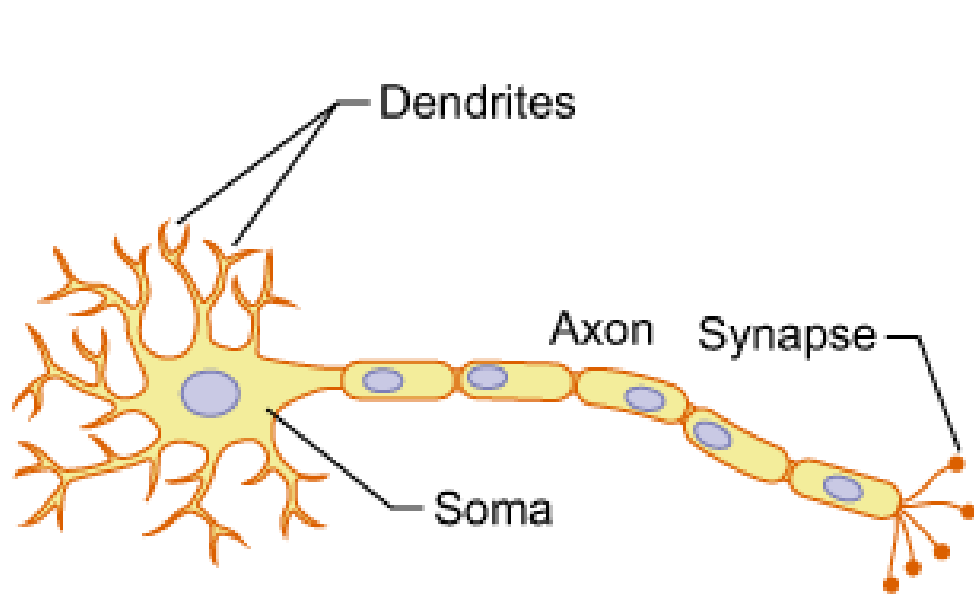
Siuly, S., Li, Y., & Zhang, Y. (2016). *EEG Signal Analysis and Classification*. Springer International Publishing. <https://doi.org/10.1007/978-3-319-47653-7>

10-10 system



Electrodes	Position
Fp1, Fp2	Frontopolar
F7, F8	Frontal (lateral)
T3, T4	Midtemporal
T5, T6	Posterior temporal
O1, O2	Occipital
Fz, Cz, Pz	Frontal, central, parietal midline
F3, F4	Midfrontal
C3, C4	Central (rolandic)
P3, P4	Parietal
A1, A2	Ear reference
T1, T2	Anterior temporal
Sp1, Sp2	Sphenoidal

Mechanism of EEG



Mechanism of EEG

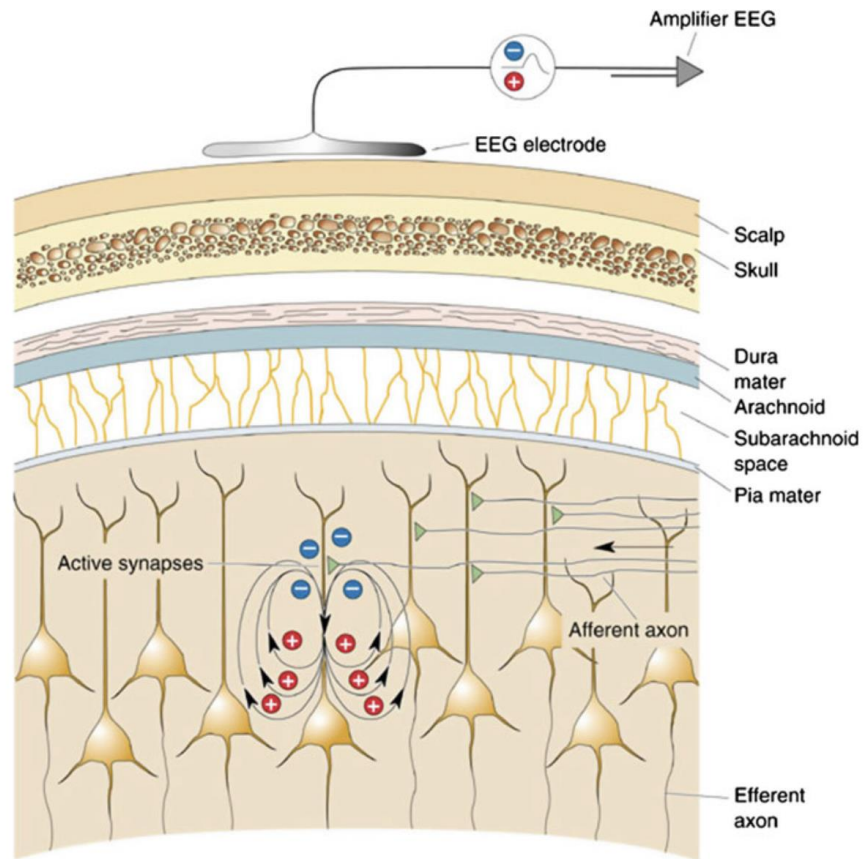


Fig. 1.9 Illustration of generation of very small electrical fields by synaptic currents in pyramidal cells. The EEG electrode measures the signal through the thick layers of tissues. Only if thousands of cells simultaneously their small voltages can the signals become large enough to be seen at the surface of the scalp (Freeman 2004a, b)

Postsynaptic potentials (PSP) by a large population of pyramidal cells (neurons) generate electrical fields that can be record by EEG electrodes on the scalp

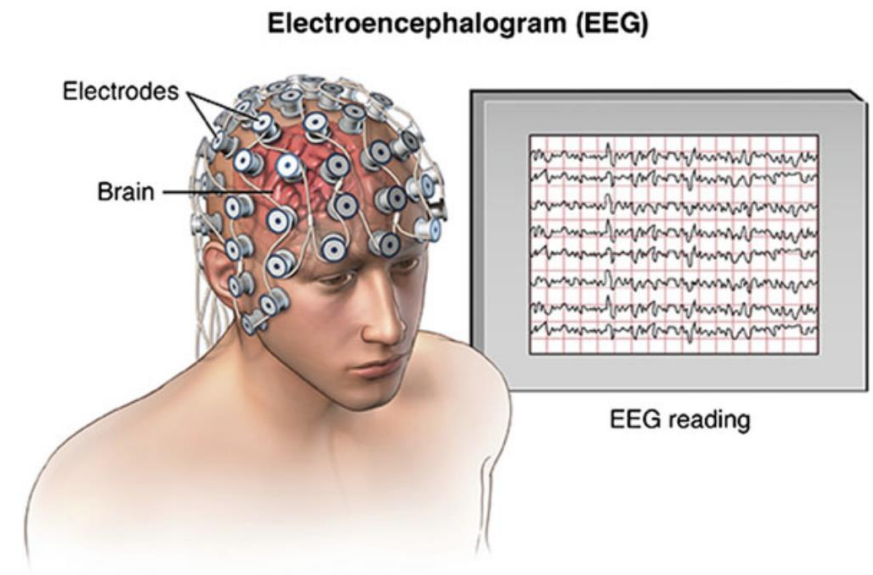


Fig. 1.1 An illustration of EEG recording (Ref. EEG, Saint Luke's Health System)

Applications of EEG

- Medical diagnosis, e.g., Epilepsy, sleep disorders
- Brain-computer interfaces (BCIs)
- Psychology
 - Decision making
 - Language processing
 - Aging

<https://www.youtube.com/watch?v=LWX6m5CsO-g>

<https://www.youtube.com/watch?v=3NRxckSlB9s>

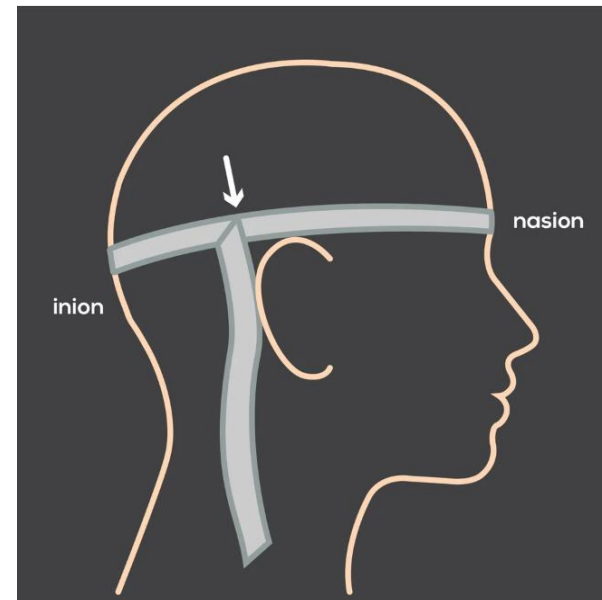
Guidelines on EEG data collection – resting-state EEG (using SAGA TMSi)

1. Connect the amplifier to the power source
2. Connect the cap, ground electrode, and external electrodes to the amplifier
3. Open the data collection software
4. Settings
 - Electrodes (including two bipolar channels)
 - Average reference
 - High-pass filter
 - Scaling if necessary

Guidelines on EEG data collection – resting-state EEG (using SAGA TMSi)

5. Measure the head size, help the participant to wear the cap
6. Make sure that the cap is centered, and the distance between the cap and eyebrows should be one tenth of the distance between the eyebrows to inion.

Size	Circumference
M	54-58 cm
L	58-62 cm



Guidelines on EEG data collection – resting-state EEG (using SAGA TMSi)

5. Measure the head size, help the participant to wear the cap
6. Make sure that the cap is centered, and the distance between the cap and eyebrows should be one tenth of the distance between the eyebrows to inion.
7. Inject gel on the scalp for each electrode position, observe impedance (should be lower than $5k\Omega$)
8. Check if all channels look normal, if not, fix them by injecting more gel. (if it doesn't work, mark the specific electrode that doesn't work well)
9. For external electrodes, wipe the skin with abrasive gel before sticking electrodes
10. Press the physical button to indicate the beginning of recording

Mental arithmetic task (serial subtraction)

4753 – 17 – 17 – 17 - ...

"Davis speculated that those with strong alpha waves might have been more relaxed than those with weak waves." (Baloh, 2024, p. 346)



Hallowell Davis (1896-1992)

EEG data – experimental design

- Resting-state EEG
- Event-related potentials (ERPs)
 - What is the typical EEG signal following an event such as stimulus or response

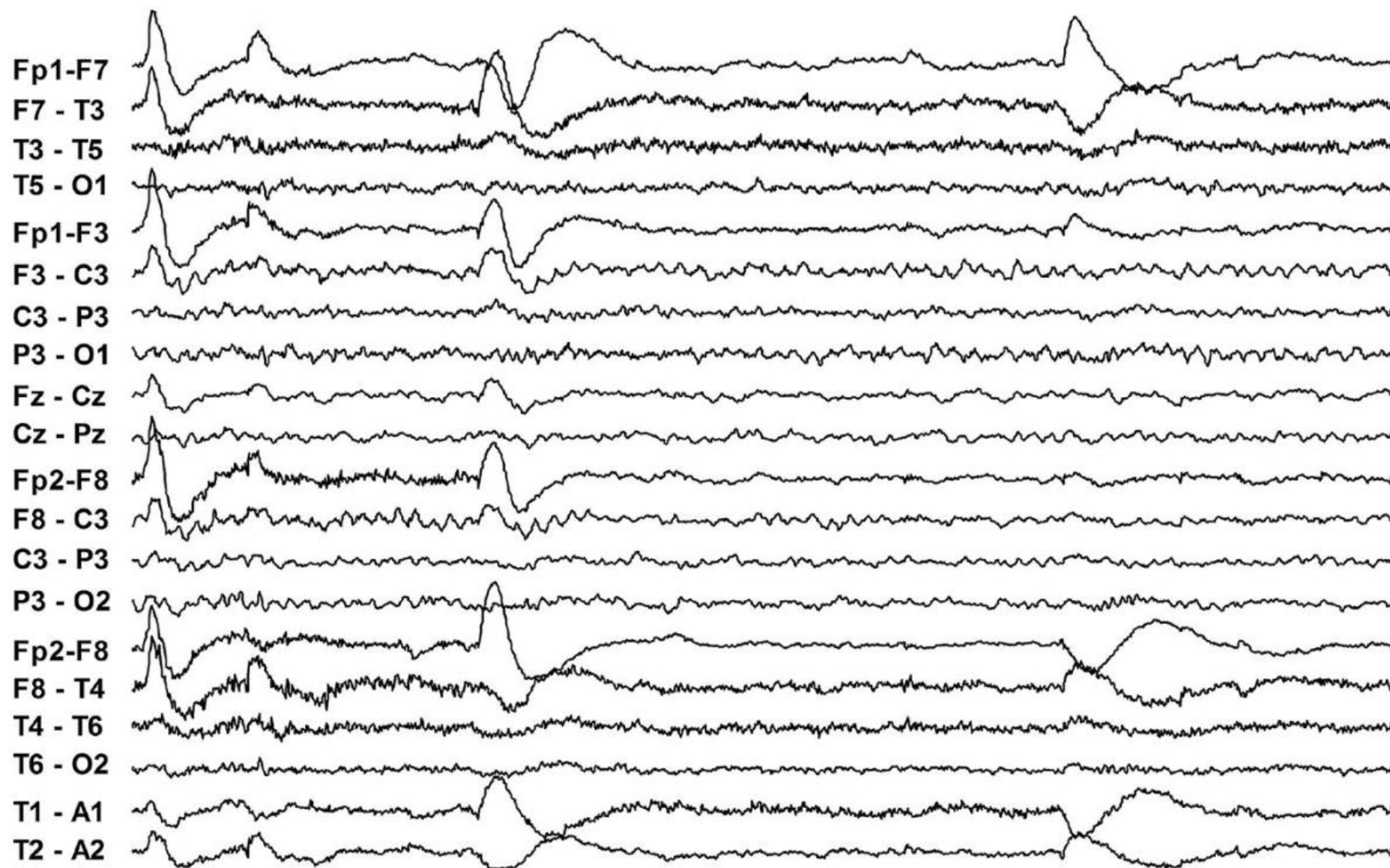
EEG data - artefacts

- Heartbeat / EKG
- Eye blinks
- Eye movement
 - Left vs. Right
 - Up vs. Down

A

look right

look left



100 μ V

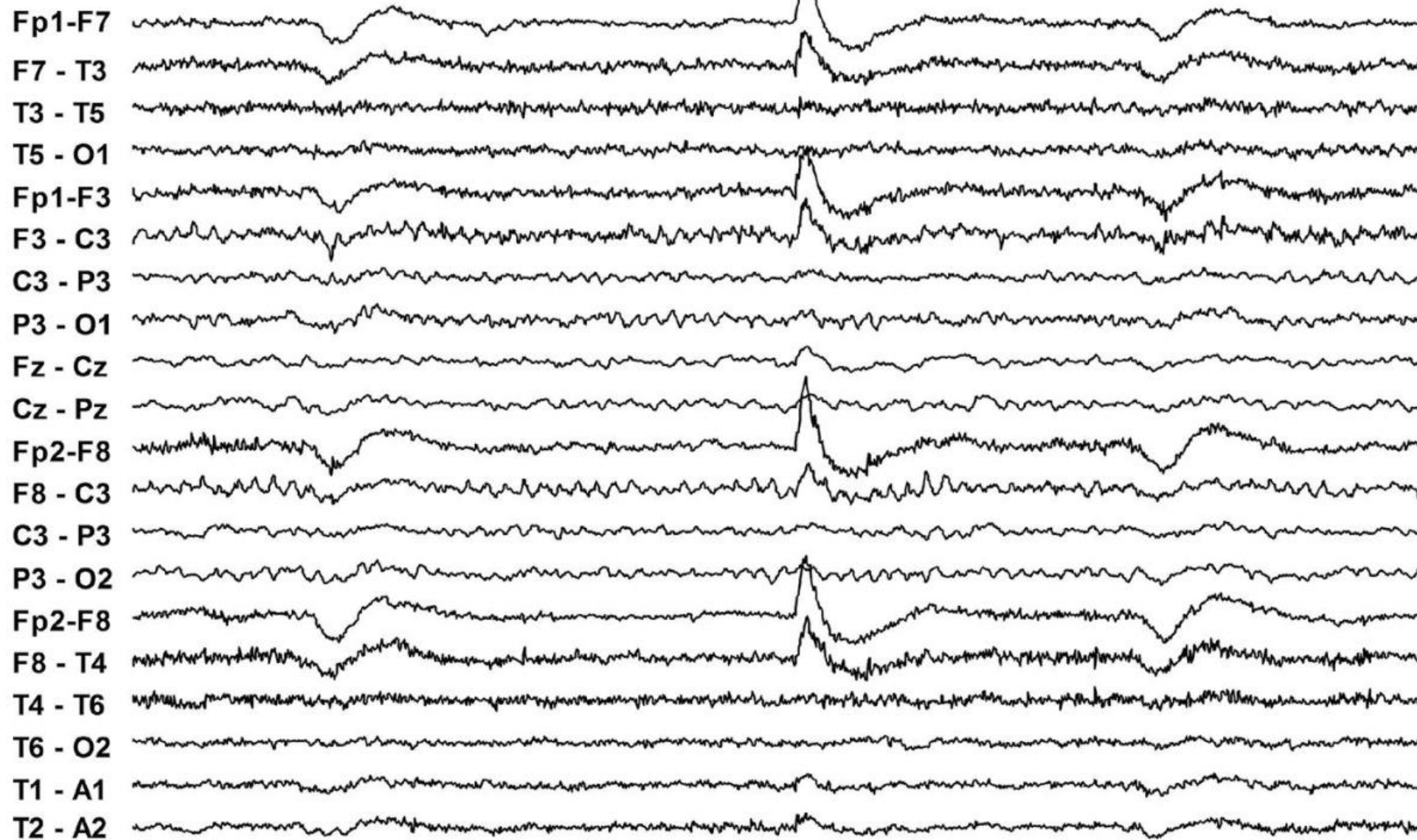
1 s

B

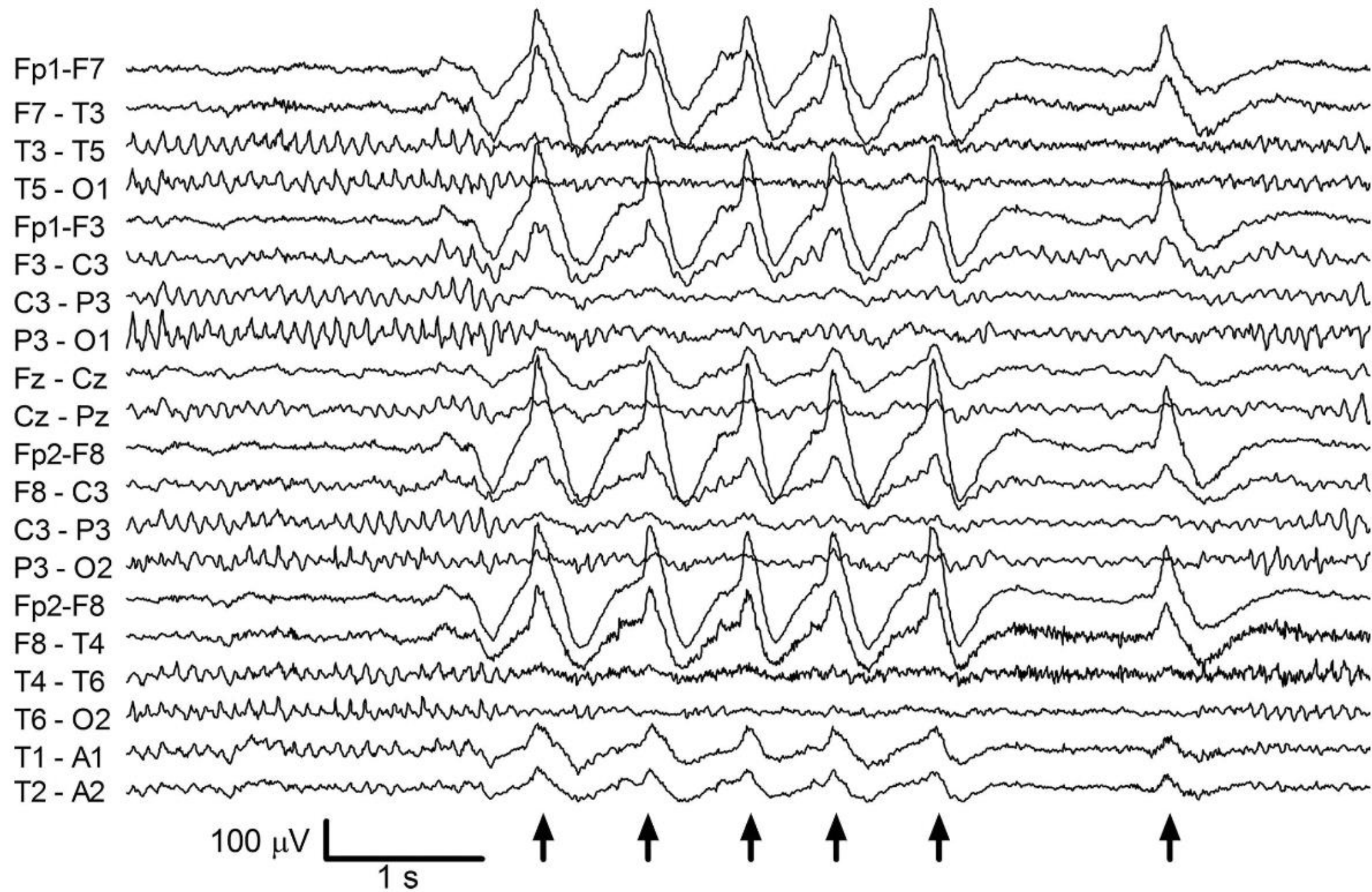
Look up

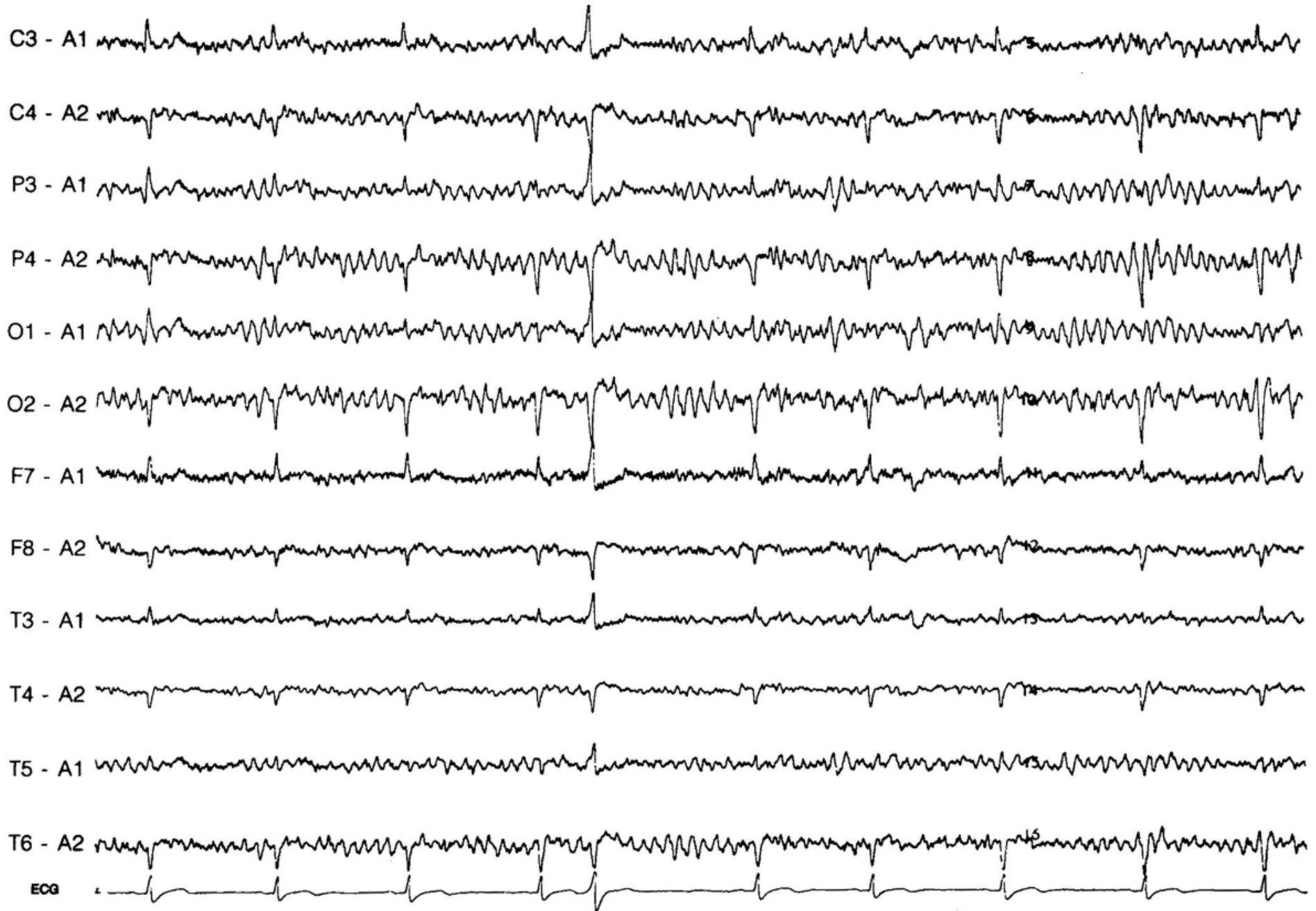
look down

look up

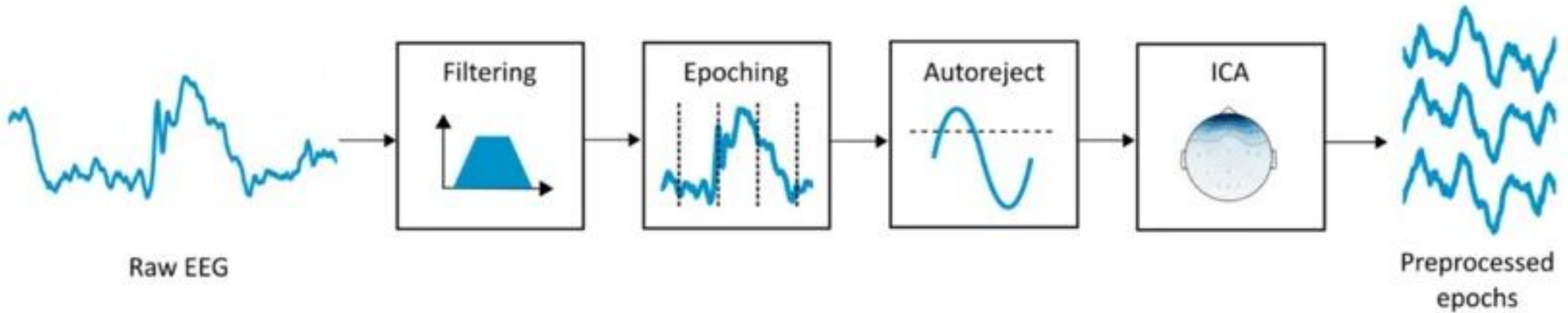


100 μ V |
1 s





EEG data - preprocessing



EEG data – Spectral analysis

- Spectral analysis (frequency power)

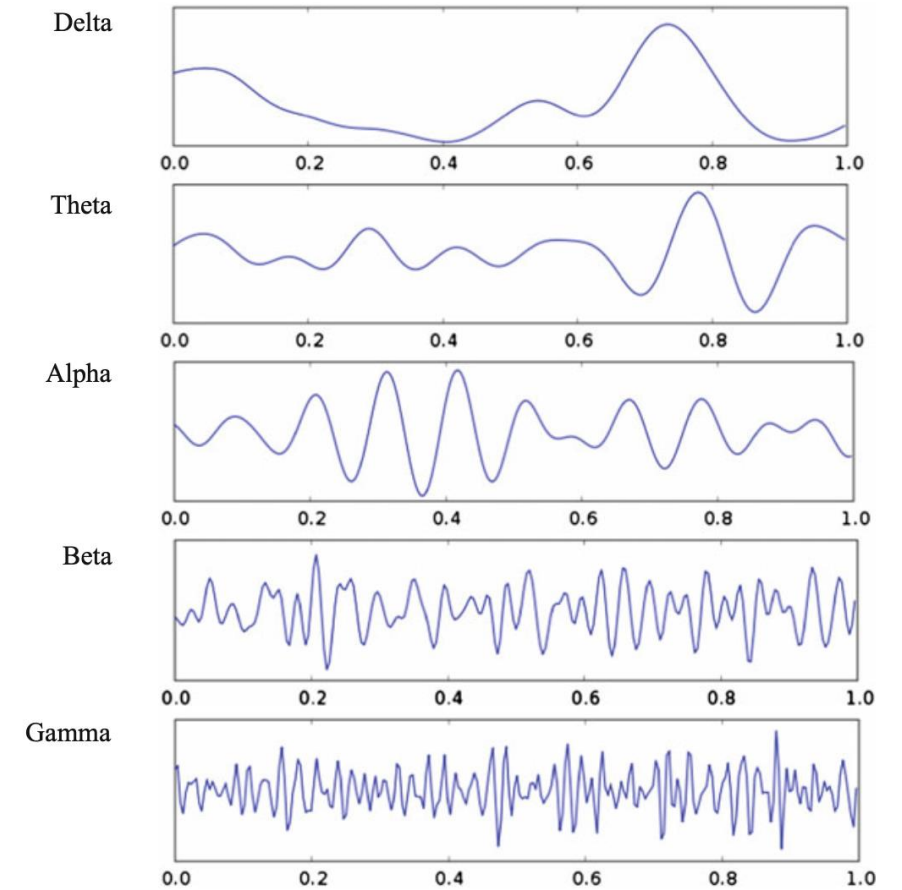
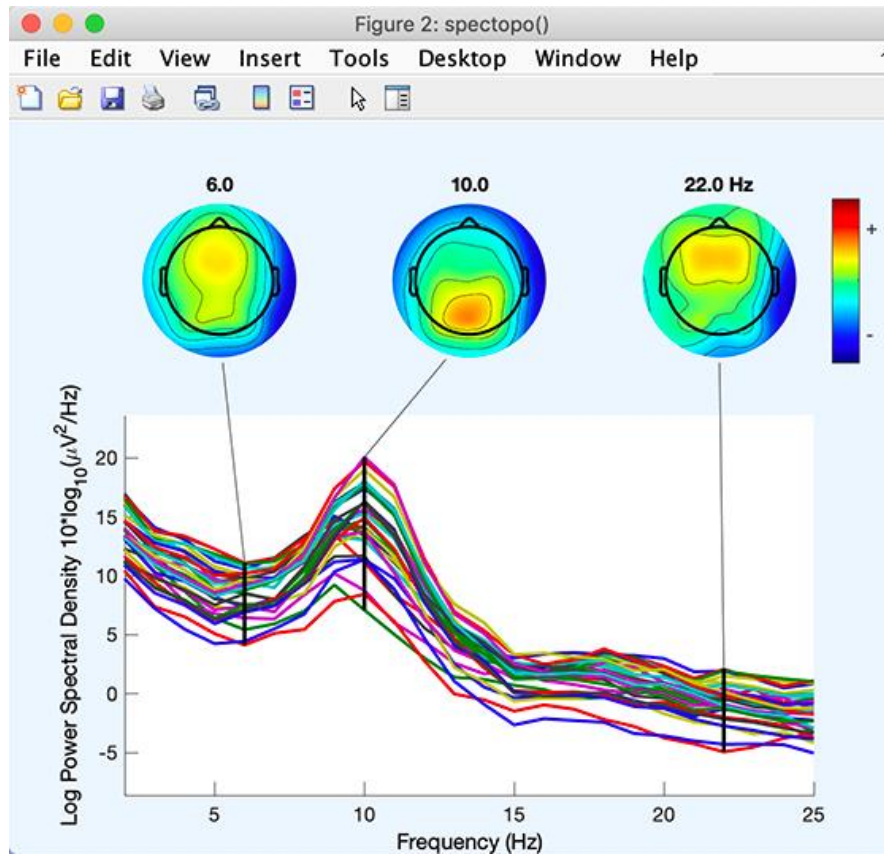
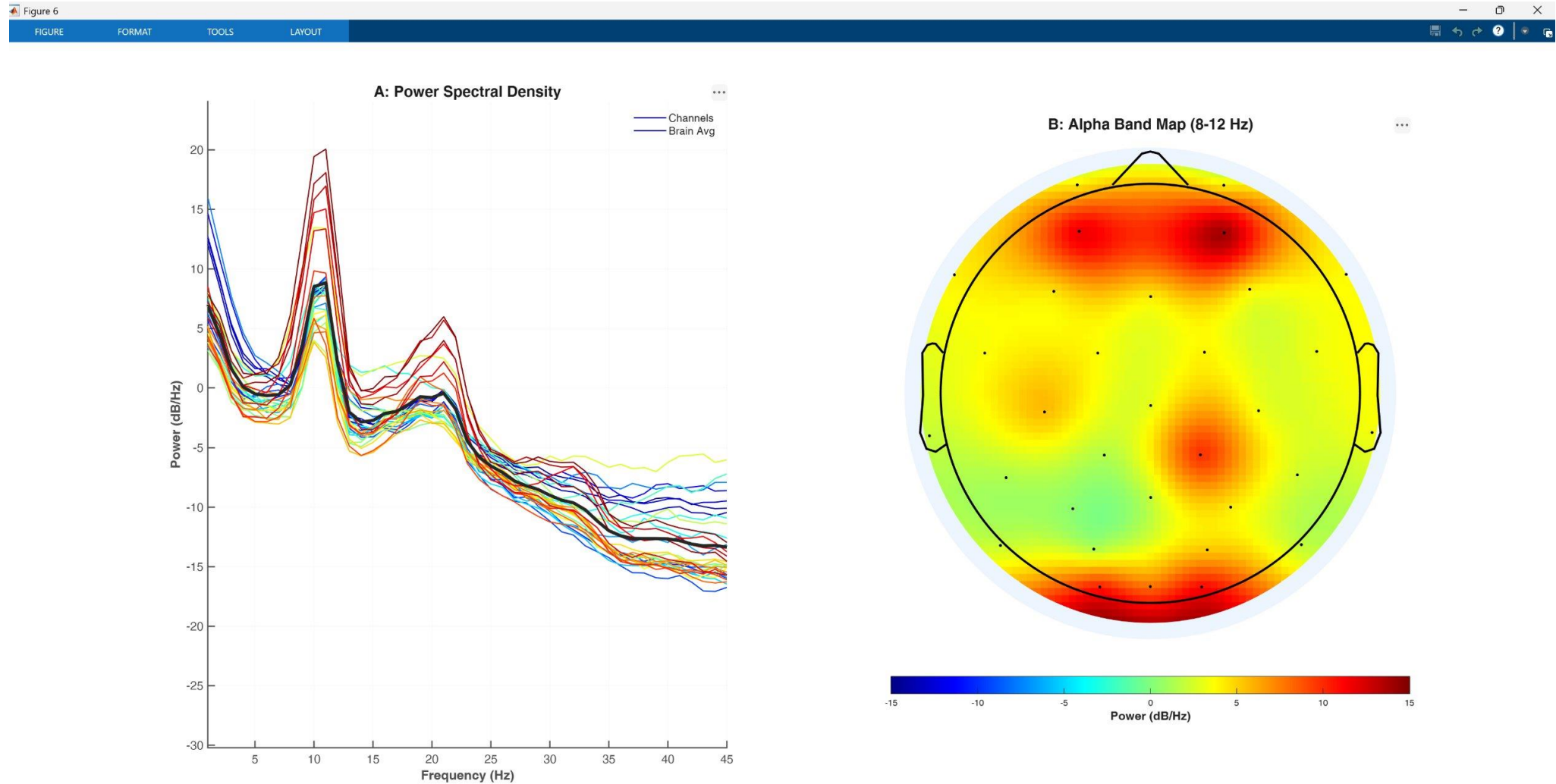
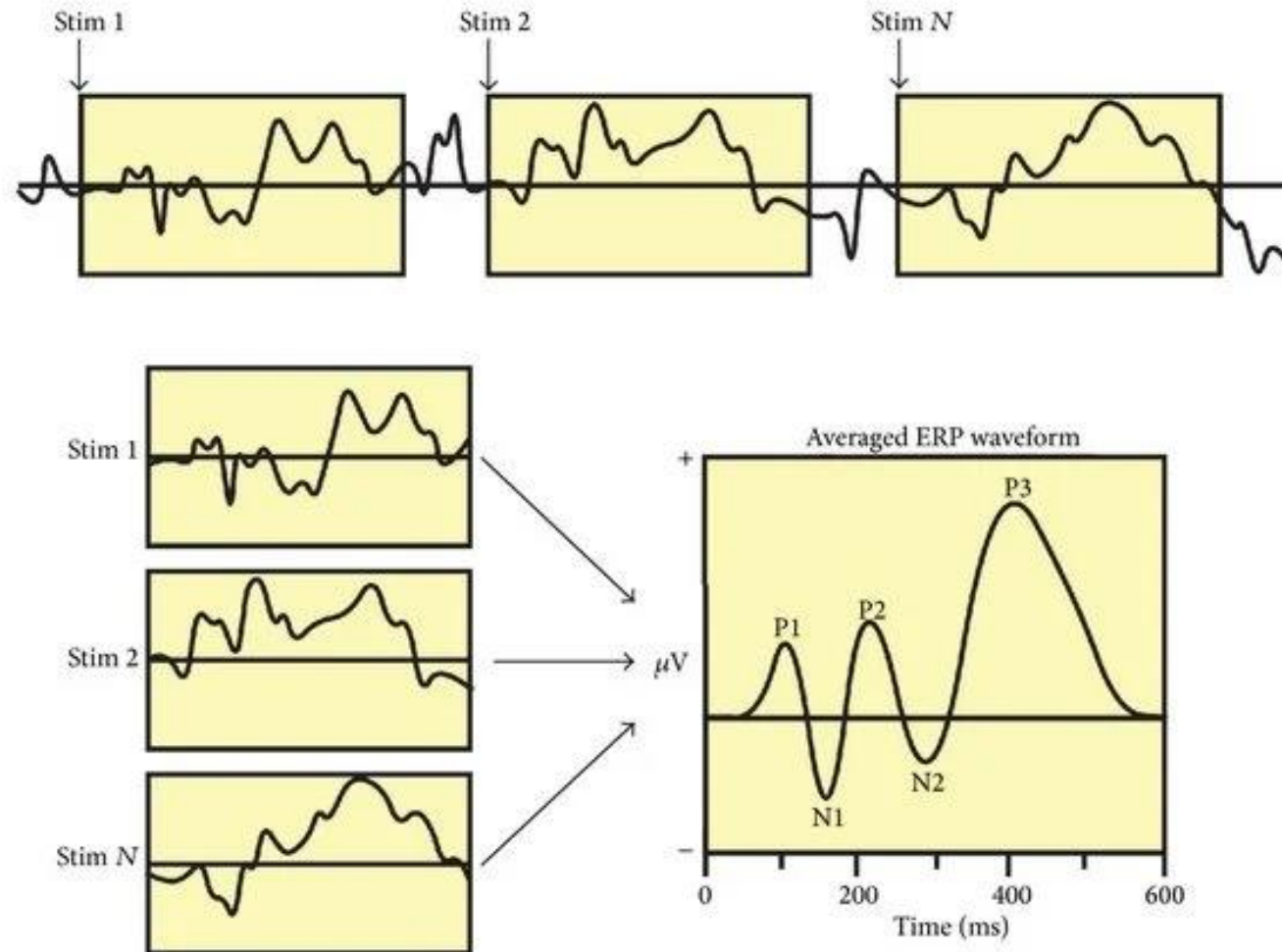


Fig. 1.11 Example of different types of normal EEG rhythms (Ref. Brainwave Entrainment, Itsu sync)

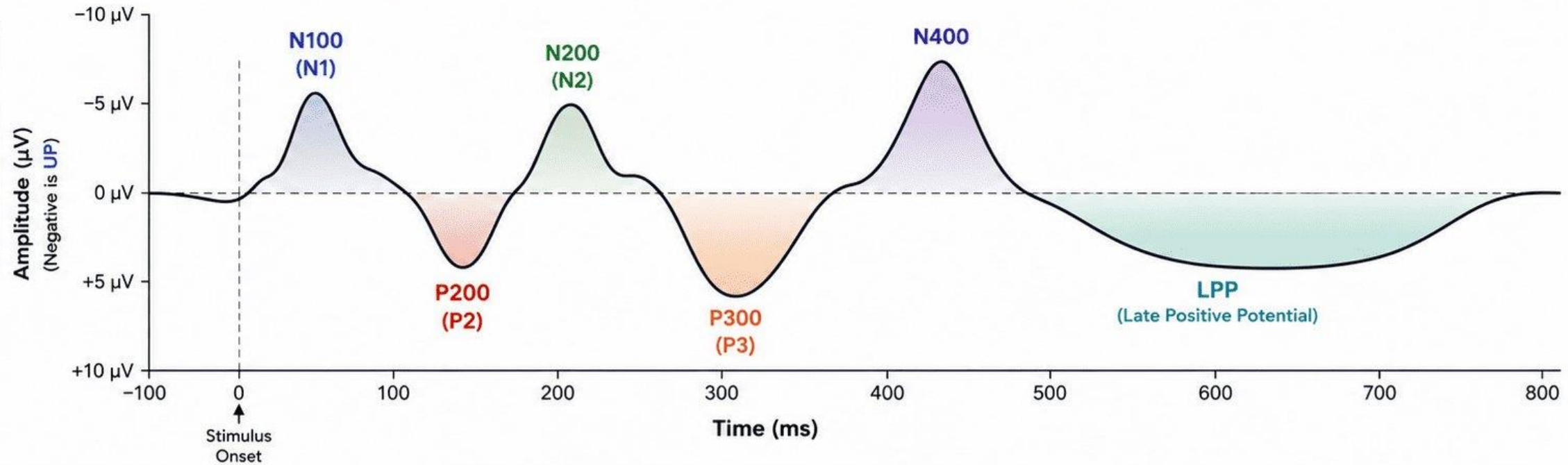
EEG data – Resting state (eyes closed)



EEG data - ERP



Major ERP Components and What They Tell Us



N100 (N1)

The Brain's First Detection



- Appears ~100 ms after stimulus
- Reflects the brain's initial sensory detection: "something happened"
- Present for sounds, visual stimuli, and touch
- Larger N1 = the brain detected the stimulus more strongly

P200 (P2)

Early Classification



- Appears ~200 ms after stimulus
- The brain begins classifying what it detected
- Modulated by attention: if you're paying attention, P2 changes

N200 (N2)

Conflict and Control



- Appears ~200 ms after stimulus
- Related to cognitive control, conflict detection, and inhibition
- Active when the brain is saying "wait, I need to override my automatic response"

P300 (P3)

The Attention and Memory Signal



- Appears ~300 ms after stimulus
 - The most studied ERP component in history
 - Reflects attention, surprise, and working memory updating
 - Larger when a stimulus is rare or unexpected
 - Used in lie detection research, ADHD diagnosis, BCI
- (Link to future Post 8 for deep dive on P300)*

N400

The Language Surprise Signal



- Appears ~400 ms, specifically for words and language
- Larger when a word is unexpected or doesn't make sense in context
- Example: "I drink my coffee with cream and socks" → huge N400 at "socks"

LPP

(Late Positive Potential)
The Emotional Attention Signal



- Appears 400–800 ms after stimulus
- Reflects sustained emotional attention
- Larger for emotional images (both positive and negative) vs neutral

Resources

- Baloh, R. W. (2024). *Brain Electricity: The Interwoven History of Electricity and Neuroscience*. Springer Nature Switzerland. <https://doi.org/10.1007/978-3-031-62994-5>
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- Op De Beeck, H., & Nakatani, C. (2025). *Introduction to Human Neuroimaging* (2nd ed.). Cambridge University Press. <https://doi.org/10.1017/9781009349925>